

Multi-criteria assessment of cropping systems for the sustainable intensification in the Pampas

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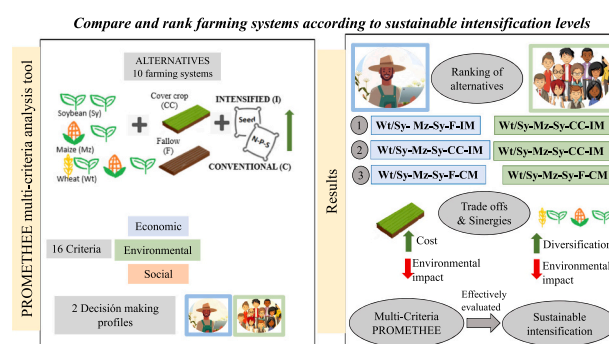
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HIGHLIGHTS

- Multi-criteria analysis helps to assess the sustainability of farming systems.
- We compare and rank farming systems according to sustainable intensification levels.
- Wheat/soybean-maize-soybean is the most preferred rotation for the Pampas.
- Costs of cover crops discourage adoption despite of the environmental benefits.
- PROMETHEE effectively evaluated the sustainable intensification in crop sequences.

GRAPHICAL ABSTRACT



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ABSTRACT

CONTEXT: The transformation of cropping systems towards sustainable intensification (SI) is essential in order to meet the demands of society. In the Pampas region of Argentina, most agricultural research has focused on crop ecophysiology and agronomic factors, although what seems to be lacking is an integrated evaluation of cropping systems to determine their alignment with SI trends. Multi-criteria analysis (MCDA) methods provide a formal framework to assess trade-offs between production and sustainability, rendering them useful for SI analysis.

OBJECTIVE: This paper aims to compare and rank the cropping systems available in the rolling Pampas, the corn belt of Argentina, based on their alignment with SI trajectories, using the multi-criteria decision model “Preference Ranking Organization Methods for Enrichment Evaluations” (PROMETHEE).

METHODS: We used information from a mid-term experiment conducted in Pergamino, Buenos Aires, Argentina, to compare ten cropping systems over five consecutive years. These systems resulted from combining three crop sequences with either fallow or cover crops during the winter, and were grown using two different management practices. Sixteen economic, socioeconomic, and environmental criteria were defined to assess each of the cropping systems. We combined PROMETHEE with the Analytical Hierarchy Process (AHP) to compare the cropping systems. We considered two different decision-making profiles: one obtained through interviews with

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farmers, and the other based on interviews with agents involved in sustainable production systems at the Ministry of Agriculture of Argentina.

RESULTS AND CONCLUSIONS: We ranked the cropping systems assessed according to how well they aligned with SI, the system that combines improving profits with reduced environmental impact. We found that the three-year rotation sequence of wheat/soybean-corn-soybean is the most aligned to the sustainable intensification option among the feasible crop sequences implemented in the rolling Pampas region. Furthermore, our results indicate that cover crops provide environmental benefits, but the associated costs discourage their implementation when the farmer profile focuses on maximizing economic benefits.

SIGNIFICANCE: We emphasize the importance of integrally analyzing cropping systems. Our findings confirm the advantages noted in prior studies and provide a more thorough evaluation of wheat/soybean-corn-soybean three-year sequence. Finally, our work demonstrates the value of PROMETHEE multi-criteria analysis in assessing the SI of cropping systems. This tool could be a useful aid in facilitating discussions among farmers, consultants and other stakeholders, in the search for sustainable intensification pathways.

1. Introduction

Society worldwide is questioning the highly simplified cropping systems due to their environmental impacts (Foley et al., 2011; Wingeyer et al., 2015), which include soil degradation, biodiversity loss, greenhouse gas emissions, nutrient imbalances, and risks of pesticide contamination (Pretty et al., 2018; Rockström et al., 2017). Society demands that cropping systems produce enough food, fiber, and fuels while also minimizing environmental impact and providing positive social and cultural contributions (Pretty et al., 2018). Therefore, the change of cropping systems towards sustainable intensification (SI) trajectories is necessary.

There is broad consensus on defining SI as “increasing production on the same area of land while reducing environmental impacts and, at the same time, increasing contributions to natural capital and the flow of environmental services from agriculture” (Pretty, 2008; Baulcombe et al., 2009; Godfray et al., 2010; Conway and Waage, 2010; Pretty et al., 2011). The notion of sustainable intensification denotes an aspiration of what is pursued by society, rather than a definition of concrete ways of how to achieve the related objectives (Garnett and Godfray, 2012). Different practices oriented towards SI have been proposed worldwide, such as integrated pest management, soil conservation practices, redesigned cropping systems to integrate production aims with biodiversity conservation schemes, the inclusion of trees in cropping systems, and irrigation water management (Pretty et al., 2018).

Although the definition of SI is still under discussion (Garnett and Godfray, 2012; Mahon et al., 2018; Pretty and Bharucha, 2014; Rockström et al., 2017), all views agree on the need to consider the multifunctionality of cropping systems (Vanclay, 2009; Pretty et al., 2018; Doré et al., 2011; Titttonell, 2014). The pillars of SI are productivity, economics, environment, and social and human well-being (Smith et al., 2017). While the criteria to assess productivity, economics, and environmental impacts are clear, the criteria for social and human well-being are less clear (Smith et al., 2017). Due to its multifunctionality, the assessment of SI in cropping systems requires more integral perspectives (Mahon et al., 2018).

The use of multi-criteria decision analysis (MCDA), such as the Preference Ranking Organization Method for Enrichment Evaluations (PROMETHEE), is an appropriate approach to assess sustainability in different cropping systems and regions. PROMETHEE allows combining objective measures with subjective information based on how particular actors perceive their importance.

The difficulty of applying the definition of SI can be explained by the fact that there are few specific suggestions on how to apply it in particular scenarios. Nonetheless, some studies have assessed SI options using insightful approaches (Mouratiadou et al., 2021; Barnes and Thomson, 2014; Firbank et al., 2013; Gadanakis et al., 2015). Multi-criteria decision analysis (MCDA) has been employed to assess sustainability in various cropping systems and regions (e.g., Evia and Sarandón, 2002; Sadok et al., 2008; Diaz-Balteiro et al., 2020; Talukder, 2016). With regards to MCDA, discrete methods are suitable for ex-post

evaluations of field experiments. The Preference Ranking Organization Method for Enrichment Evaluations (PROMETHEE) is a discrete multi-criteria decision technique based on an over-classification model. PROMETHEE has been applied in various studies related to agriculture. For instance, Kyllili et al. (2016) utilized this technique to determine the optimal energy crops. Król et al. (2018) used the methodology to evaluate and rank climate change adaptation practices in maize cropping systems related to tillage. Talukder and Hipel (2018) used the method to compare the sustainability of five cropping systems in coastal Bangladesh. In this paper, we propose PROMETHEE as a tool to operationalize SI concepts. PROMETHEE allows for the combination of objective measures with subjective information based on how particular actors perceive their importance (Sadok et al., 2008).

In the Pampas region of Argentina, some cropping systems have been studied in mid- and long-term experiments. Ex-post analyses of these trials generally consider some aspects of the cropping systems, such as resource use, efficiencies, nutrient balance, deep drainage, nitrogen leaching, carbon sequestration and organic matter of soils (Andrade et al., 2015; Andrade et al., 2017; Portela et al., 2016; Restovich et al., 2019; Martínez et al., 2020; Crespo et al., 2021). However, so far, no research has been carried out to compare cropping systems using a more comprehensive methodology based on multiple criteria. Hence, our analytical approach incorporates economic, social and environmental dimensions, as well as information on how stakeholders perceive the relevance of these dimensions, in order to identify the most compatible SI trajectories. The lack of an integrated perspective may be due to the difficulty of conducting multi-criteria comparisons, given the multiple dimensions of the SI concept and the potential trade-offs among criteria (Mouratiadou et al., 2021).

While the SI concepts are clearly presented in the aims, the rules that define the extent of alignment between productive systems and these concepts are vague (Garnett and Godfray, 2012; Pretty et al., 2018). The trajectory of SI should also be assessed at the regional level, considering the differences introduced by the diverse natural environments and socioeconomic factors (Weltin et al., 2018). To establish sustainable intensification trajectories, it is crucial to identify alternatives that enhance productivity, generate better economic outcomes, and have a reduced negative impact on the environment. The aim of this article is to compare and classify cropping systems in the rolling Pampas of Argentina based on their alignment with the concept of SI. Previous studies have shown that the most diversified crop sequences are the most suitable from an ecophysiological perspective in the Argentine Pampas region. Our objective is to demonstrate that, as the concept of SI implies, the most diversified production systems are also the best options from an integrated viewpoint, considering economic, social, and environmental factors.

2. Materials and methods

The study is based on information obtained from a field experiment conducted in a farm located in Pergamino (33° 51' 55.7" S; 60° 40' 28.39

W), which is a representative area of the rolling Pampas, the corn belt of Argentina. The experiment was a split plot factorial including five cropping systems (subplots), each implemented under fallow or with cover cropping, arranged in a 3-block design, according to topography and soil organic matter content (Hisse et al., 2022). The soil is a deep Mollisol, Typic Argiudoll with approximately 3% organic matter in the topsoil. The climate is humid subtropical, characterized by hot summers and no marked dry season. The mean annual rainfall is 980 mm and the mean annual temperature is 18 °C (1938–2019). The study period covered five years, from 2015 to 16 to 2019–20, during which the average rainfall was 1131 mm (2015–16: 1434 mm, 2016–17: 1138 mm, 2017–18: 896 mm, 2018–19: 1315 mm, and 2019–20: 874 mm).

The study assessed ten different cropping systems over a period of five growing seasons, from 2015 to 2016 to 2019–2020. These cropping systems were created by combining three different factors: winter period management, crop sequences, and agronomic management (Table 1). The winter period was either left fallow or sown with cover crops, while the crop sequences included soybean mono-cropping (Sy-Sy), a two-year crop sequence alternating maize and soybean (Mz-Sy), and a three-year crop sequence including the wheat/soybean double crop and single crops of maize and soybean (Wt/Sy-Mz-Sy). Two cropping systems were assessed: conventional management (CM) and intensified management (IM). The Mz-Sy and Wt/Sy-Mz-Sy crop sequences were assessed under both cropping systems, while soybean mono-cropping was only assessed under the CM because it is considered unsustainable and highly simplified (Caride et al., 2013).

The cover crops used in this study were a mixture of triticale (*Triticosecale Wittm. ex A. camus*) and hairy vetch (*Vicia villosa* Roth). The CM included the most commonly used crop management practices recommended to farmers in the region. The IM, on the other hand, was based on prior knowledge aimed to increase grain yields by improving resource productivity (Cassman, 1999). The main differences between the CM and IM were in the choice of genotypes, plant density, and crop nutrition and protection schemes. The IM used top-quality inputs such as genotypes, agrochemicals and fertilizers and the fertilization levels of nitrogen, phosphorus, and sulfur were relatively higher than in the CM (Annexes E and F). Both agronomic management used no-tillage because it is widely used in the region. This field experiment had the advantage of including all crop types each year, reducing the possible effects of interannual climatic variability. Details of the experiments, as well as the statistical analysis, are presented in Hisse et al. (2022).

2.1. Criteria assessed in the field experiment during five growing seasons

Sixteen criteria were defined to assess each cropping system, based on the SI conditions described in the literature and the main characteristics of the region.

Table 1

Ten cropping systems achieved through the integration of crop sequences, winter period management, and agronomic management.

Crop sequence	Winter period	Agronomic management	Cropping systems
Soybean	Fallow	Conventional	Sy-F-CM
Maize-Soybean	Fallow	Conventional	MzSy-F-CM
Maize-Soybean	Fallow	Intensified	MzSy-F-IM
Wheat/Soybean-Maize-Soybean	Fallow	Conventional	Wt/SyMzSy-F-CM
Wheat/Soybean-Maize-Soybean	Fallow	Intensified	Wt/SyMzSy-F-IM
Soybean	Cover crops	Conventional	Sy-CC-CM
Maize-Soybean	Cover crops	Conventional	MzSy-CC-CM
Maize-Soybean	Cover crops	Intensified	MzSy-CC-IM
Wheat/Soybean-Maize-Soybean	Cover crops	Conventional	Wt/SyMzSy-CC-CM
Wheat/Soybean-Maize-Soybean	Cover crops	Intensified	Wt/SyMzSy-CC-IM

The economic criteria included gross margin, return on working capital, and economic risk. Gross margin is the most frequently used indicator to assess the economic performance of annual crops (Ghida Daza, 2009). Return on working capital and economic risk have been used to compare cropping systems (Pecar, 2008).

The amount of export tax payments was considered a socioeconomic criterion, as these payments represent additional income in terms of tax collection that could be used to improve income distribution and social welfare (Rzezak, 2008).

The environmental criteria were identified as indicators of the impact of agricultural production systems on the environment in the Argentine Pampas: nitrogen, phosphorus, and sulfur balances (García and González Sanjuan, 2010), soil carbon balance (Berhongaray et al., 2013), net greenhouse gas emissions (Lewczuk, 2017; Caviglia et al., 2016), environmental impact due to pesticide use (Bernardos and Zaccagnini, 2011), and diversity and intensification of crop sequences (Caviglia and Andrade, 2010). The criteria are described in Table 2, and details of the parameters used in these computations can be found in Annex A.

In the multicriteria analysis, the positive and negative soil nitrogen and sulfur balances were considered as separate indicators. This is because the magnitude of positive nitrogen and sulfur balances of the soil is closely related to the amount of nutrient that is leached (Hatch et al., 2003).

2.2. Multi-criteria analysis

The PROMETHEE discrete multi-criteria analysis method (Brans and Vincke, 1985) was employed to compare the outcome of the ten cropping systems assessed in the experiment. This method involves pairwise comparisons of the cropping systems using a differential function for each sustainable intensification criterion.

$$d_j(A_h - A_k) = C_j(A_h) - C_j(A_k) \quad (1)$$

where d_j represents the difference between the values h and k of two cropping systems (A) for the j criterion (C).

For each criterion, the V-shape preference function was defined (Brans and Vincke, 1985; Brans et al., 1986).

$$P_j(A_h - A_k) = f(d_j(A_h - A_k)) \quad (2)$$

where P_j denotes the preference between the values of cropping systems h and k for criterion j .

The V-shape function is defined as follows:

$$f(d_j(A_h - A_k)) = 0 \text{ if } d_j(A_h - A_k) \leq 0 \quad (3)$$

$$f(d_j(A_h - A_k)) = (d_j(A_h - A_k))/p \text{ if } 0 \leq d_j(A_h - A_k) \leq p \quad (4)$$

$$f(d_j(A_h - A_k)) = 1 \text{ if } d_j(A_h - A_k) > p \quad (5)$$

where p represents the threshold value obtained by calculating the difference between the maximum and minimum values for each criterion. This preference function serves to normalize the differences in values for the various criteria, ranging from 0 to 1.

Next, the multicriteria preference index is calculated

$$\pi(A_h, A_k) = \sum_{j=1}^n P_j(A_h, A_k) W_j \quad (6)$$

where π denotes the preference index between the values of the cropping systems h and k and W_j represents the weight assigned to each criterion j .

Using the multicriteria preference index, we can obtain the positive outperforming flow $\phi^+(A_h)$ and negative outperforming flow $\phi^-(A_h)$ for each cropping system. The positive outperforming flow indicates how much a cropping system is outperforming the other alternatives, with a

Table 2

Description of the economic, socioeconomic, and environmental criteria employed in the analyses.

	Criteria	Description
Economic	Gross margin	Gross income - direct costs (expenditure on inputs, services, harvesting, and marketing) (Ghida Daza, 2009).
	Economic risk	Minimum annual gross margin in the 5-year period.
	Return on working capital	Gross margin divided by the cost of inputs and services.
Socioeconomic	Amount of export tax payments	Export tax rate which is specific to each product
	Soil carbon balance	AMG model of the evolution of soil organic carbon (SOC): Stable carbon stock + mass of carbon contributed by the crop * humification coefficient/mineralization coefficient. (Andriulo et al., 1999; Milesi Delaye et al., 2013).
	Net greenhouse gas emissions	Carbon-equivalent emission or capture: Soil carbon emissions or removals according to negative or positive soil carbon balance + N ₂ O emissions from nitrogen fertilization + N ₂ O emissions from mineralization + CO ₂ emissions from urea fertilization + CO ₂ , CH ₄ and N ₂ O emissions from fossil fuel use in machinery passes in the field + N ₂ O emissions from stubble input were considered for the calculation. 2006 and 2019 IPCC guidelines (Eggleston et al., 2006, Shukla et al., 2019).
Environmental	Soil phosphorus balance	P input (fertilizers) — P output (harvested grain)
	Soil sulfur balance	S inputs (fertilizers) + S input (rainfall) — S output (harvested grain)
	Soil nitrogen balance ¹	N inputs (fertilizers) + N biological fixation + N input (rainfall) — N output (harvested grain) - N output (gaseous losses).
	Environmental impact due to pesticide use - Consumer	Exposure to the product is considered through the possible presence of residues in soil and plant and the potential effect on groundwater. Kovach et al. (1992)
	Environmental impact due to pesticide use - Ecological	Comprises the effects on water and land, as well as on fish, birds, and beneficial arthropods, such as the honeybee. Kovach et al. (1992)
	Environmental impact due to pesticide use - Farm worker	Exposure to agrochemical products during application is considered by dermal toxicity and long-term exposure by chronic toxicity calculated in mammals. Kovach et al. (1992)
	Intensification sequence index	The ratio between the number of days with growing crops (from emergence to physiological maturity) and the total days of the crop season (Rodríguez et al., 2020).
	Diversity	10.000 - Herfindahl-Hirschman concentration index (HH) (Parkin and Loria, 2010). HH is computed as the sum of the squared percentages of the area assigned to each crop in the rotation. (Kumar and Kumar, 2018; Ponce et al., 2015)

¹ A simplified calculation of nitrogen balance was employed, excluding losses caused by erosion, leaching, and runoff due to their high variability, which

makes estimating them challenging (Vicente and Engler, 2008; Oenema et al., 2003; Alvarez et al., 2014).

higher value indicating better performance. Conversely, the negative outperforming flow indicates how much a cropping system is being outperformed by the other alternatives, with a lower value indicating better performance.

$$\varnothing^+(A_h) = \frac{1}{m-1} \sum_{k=1}^m \pi(A_h, A_k) \quad (7)$$

$$\varnothing^-(A_h) = \frac{1}{m-1} \sum_{k=1}^m \pi(A_k, A_h) \quad (8)$$

The PROMETHEE order was determined by calculating the net out-ranking flow of each cropping system $\varnothing(A_h)$, which is the difference between its positive and negative flows, according to the equation:

$$\varnothing(A_h) = \varnothing^+(A_h) - \varnothing^-(A_h) \quad (9)$$

The Geometric Analysis for Interactive Aid (GAIA) method was used to visually display the cropping systems and criteria on a biplot graph for interactive assistance.

2.3. Weighting of criteria

To assign weights to the criteria (W_j), we utilized the Analytical Hierarchical Process (AHP), which was proposed by Macharis et al. (2004). This was necessary as the PROMETHEE methodology does not have a specific procedure for assigning weights. The AHP methodology enables the ranking of cropping systems based on the decision-making center's preferences.

Interviews were conducted to determine the importance of each criterion relative to the others. Two decision profiles were obtained from these interviews. The production-oriented decision profile arose from a joint interview with a farmer, the production manager, and the agronomic advisor of a farm in the study region. The farmer is a member of CREA (Consortios Regionales de Experimentación Agrícola <http://www.crea.org.ar>), a non-profit civil association comprised of agricultural entrepreneurs who join to share experiences and knowledge. The environmental-oriented decision profile emerged from a joint interview with members of the Direction of Sustainable Productions of the Ministry of Agriculture, Livestock and Fisheries of Argentina (MAGYP). The weight vectors obtained from these interviews are not intended to be representative of all decision profiles in society or all farmers, but rather to showcase two real relevant decision profiles that participate in different ways in the sector's decisions.

3. Results

3.1. Weight vectors

The weight vectors obtained from the interviews indicate that the environmental decision profile placed 55% more emphasis on environmental indicators compared to the productive profile, while the productive profile gave 35% more importance to economic indicators (Table 3).

3.2. Decision matrix

The decision matrix used in the PROMETHEE model was based on the objective assessment of each criterion, as presented in Table 4. The highest gross margin was achieved in the Wt/Sy-Mz-Sy-F-IM system (Table 4-A). For each crop sequence, the cropping systems managed with fallow had higher gross margin than those that included cover crops (Table 4-A). Additionally, the intensified management systems had a higher gross margin than conventionally managed systems. The economic risk was lower in the Wt/Sy-Mz-Sy-F-CM system, which also

Table 3

Target direction and weight assigned to each criterion in the PROMETHEE methodology.

Category	Criteria	Target direction	Environmental-oriented decision profile		Production-oriented decision profile	
			Weight Category	Weight Criterion	Weight Category	Weight Criterion
Economics	Gross margin	↑		0.48		3.46
	Economic risk	↓	3.79	0.41	5.88	1.73
	Return on working capital	↑		2.90		0.69
Socioeconomics	Amount of export tax payments	↑	0.32	0.32	0.38	0.38
	Soil carbon balance	↑		0.97		1.15
	Net greenhouse gas emissions	↓		1		0.49
Environmental	Soil phosphorus balance	↑		0.48		1.15
	Positive soil sulfur balance	↓		0.40		0.38
	Positive soil nitrogen balance	↓		0.50		0.49
	Negative soil sulfur balance	↓	11.88	0.40	7.70	0
	Negative soil nitrogen balance	↓		1		0
	Environmental impact due to pesticide use - Consumer	↓		0.50		0.58
	Environmental impact due to pesticide use - Ecological	↓		0.50		0.58
	Environmental impact due to pesticide use - Farm worker	↓		0.50		0.58
	Intensification sequence index	↑		2.90		1.15
	Diversity	↑		2.90		1.15

The target direction is depicted using arrows to indicate the direction in which each criterion is considered favorable in PROMETHEE. For example, a higher gross margin is regarded as desirable.

Table 4

(Panel A). Results of the criteria evaluated for each cropping system across the five cropping seasons (2015/2016; 2016/2017; 2017/2018; 2018/2019; and 2019/2020).

Cropping systems	Gross margin US\$ ha ⁻¹	Economic risk US\$ ha ⁻¹	Return to working capital %	Amount of export tax payments US\$ ha ⁻¹	Soil carbon balance tn C ha ⁻¹	Net greenhouse gas emissions kg CO ₂ eq ha ⁻¹	Diversity of sequences
Sy-F-CM	1872	184	132	1228	-1.5	7625	0
Sy-CC-CM	717	31	38	1005	1.7	-2826	5000
MzSy-F-CM	2541	317	131	739	-0.7	6799	5000
MzSy-F-IM	2570	279	98	761	-0.6	6792	5000
MzSy-CC-CM	1094	-1	43	656	1.9	-1333	6250
MzSy-CC-IM	1321	8	40	732	2.7	-3183	6250
Wt/SyMzSy-F-CM	2708	397	137	855	-0.2	4817	7500
Wt/SyMzSy-F-IM	2781	393	104	930	0.0	3809	7500
Wt/SyMzSy-CC-CM	1779	214	74	830	1.1	894	7777
Wt/SyMzSy-CC-IM	2157	258	70	939	2.1	-1935	7777

(Panel B). Results of the criteria evaluated for each cropping system across the five cropping seasons (2015/2016; 2016/2017; 2017/2018; 2018/2019; and 2019/2020)

Cropping systems	Soil phosphorus balance kg P ha ⁻¹	Soil nitrogen balance kg N ha ⁻¹		Soil sulfur balance kg S ha ⁻¹		Environmental impact due to pesticide use			Intensi-fication sequence index
		Positive	Negative	Positive	Negative	Consumer	Ecological	Worker	
Sy-F-CM	4.7	0	-148.4	3.7	-20.9	147	1364	414	0.3
Sy-CC-CM	12.2	132.5	-23.8	6.2	-15.0	123	1115	341	0.7
MzSy-F-CM	-33.3	0.5	-102.2	3.3	-12.1	159	1440	403	0.3
MzSy-F-IM	32.1	2.8	-92.8	80.5	0	141	1257	359	0.3
MzSy-CC-CM	-17.5	238.6	-12.0	5.4	-8.7	135	1227	340	0.7
MzSy-CC-IM	50.1	249.0	0	84.0	0	127	1171	318	0.7
Wt/SyMzSy-F-CM	-24.8	0	-123.4	5.1	-16.4	116	1133	314	0.4
Wt/SyMzSy-F-IM	32.0	19.8	-97.4	82.8	0	120	1071	302	0.4
Wt/SyMzSy-CC-CM	-16.8	104.4	-30.9	3.6	-12.6	114	1046	287	0.7
Wt/SyMzSy-CC-IM	34.5	152.8	-15.2	82.8	0	104	966	258	0.7

Note: Sy: Soybean; MzSy: Maize-Soybean; Wt/SyMzSy: Wheat/Soybean-Maize-Soybean; F: Fallow; CC: Cover crops; IM: Intensified Management; CM: Conventional Management.

Note: The economic risk was assessed as the minimum margin observed during the analysis period. The rate of return, diversity of sequences, and the intensification sequence index were calculated for the entire five-year period. All other criteria were calculated as the sum of the values obtained over the five analyzed crop-years.

Note: Sy: Soybean; MzSy: Maize-Soybean; Wt/SyMzSy: Wheat/Soybean-Maize-Soybean; F: Fallow; CC: Cover crops; IM: Intensified Management; CM: Conventional Management.

Note: The economic risk was assessed as the minimum margin observed during the analysis period. The rate of return, diversity, and the intensification sequence index were calculated for the entire five-year period. All other criteria were calculated as the sum of the values obtained over the five analyzed crop-years.

had the highest rate of return. The soybean mono-cropping system had the greatest contribution to export tax payments. The Mz-Sy-CC-IM system had the highest carbon balance (Table 4-A). Indeed, cover crops significantly enhanced carbon balance, and reduced net greenhouse gas emissions (Table 4-A).

Concerning the phosphorus balance, the IM systems presented positive values, while the CM systems had negative balances (Table 4-B). The systems managed with winter fallow periods had nitrogen balances closer to neutrality, while the exportation of soil nitrogen was higher. The IM systems had positive soil sulfur due to sulfur fertilization (Table 4-B). Wt/Sy-Mz-Sy-IM showed the best environmental impact results regarding pesticide use (consumer, farm worker, and ecosystem), and overall, the IM systems had lower toxicity (Table 4-B). The systems with cover crops outperformed those with fallow periods in the intensification sequence index. Finally, the highest diversity was observed in systems including the most diverse crop sequence Wt/Sy-Mz-Sy (Table 4-B).

3.3. Ranking of cropping systems in PROMETHEE

The cropping systems were ranked using the PROMETHEE method for both the production-oriented and environmental-oriented decision profiles. For the production-oriented profile, the crop sequence Wt/Sy-Mz-Sy was ranked in the top three positions, with IM outperforming CM within this sequence. Systems with fallow were ranked higher than those with cover crops, despite the small difference between the two winter period managements (Fig. 1- Panel A). For the environmental-oriented profile, the cropping system Wt/Sy-Mz-Sy-CC was ranked in the first and second places, with IM outperforming CM within this sequence as well. (Fig. 1- Panel B). Notably, the Wt/Sy-Mz-Sy-CC-IM system was ranked second in the production-oriented profile and rose to first place in the environmental-oriented profile, with a very short distance from the first cropping system. The sensitivity analysis is presented in Annex B.

3.4. Effect of each criterion on the performance of cropping systems

Performance by criterion is shown for each cropping system (Figs. 2 and 3). These bar graphs combine the information of the criteria indicator values (Table 4) and weight of the stakeholders' decision profiles (Table 3), because the flows are determined based on the value obtained for each criterion and the assigned weight.

Particularly, it is interesting to analyze the criteria contributions to the performance of the top-ranked cropping systems. For the production-oriented decision profile, the first cropping system Wt/Sy-Mz-Sy-F-IM has gross margin and economic risk as the main criteria in the positive flow, which are the criteria with the greatest weight for this profile. In the second cropping system Wt/Sy-Mz-Sy-CC-IM, the main criteria in the positive flow are the intensification sequence index and soil carbon balance. Both cropping systems have a positive soil sulfur balance in the negative flow, indicating that an adjustment in sulfur application could enhance their performance. Wt/Sy-Mz-Sy-F-IM system also has net greenhouse gas emissions, soil carbon balance and intensification sequence index in the negative flow, whereas Wt/Sy-Mz-Sy-CC-IM has return on working capital and a positive soil nitrogen balance. This difference is associated with the inclusion or not of cover crops in the sequences (Fig. 2).

For environmental-oriented decision profile, the first two cropping systems (Wt/Sy-Mz-Sy-CC-I and Wt/Sy-Mz-Sy-CC-C) have the intensification and diversity indices of crop sequences as main criteria in the positive flow. In the negative flow, both cropping systems have the surplus of nitrogen and the return on working capital as criteria. In the first cropping system (Wt/Sy-Mz-Sy-CC-I), a positive soil sulfur balance is also added to the negative flow. An adjustment in nitrogen and sulfur management could increase the positive flow and, consequently, improve the rate of return on working capital (Fig. 3). In the second cropping system (Wt/Sy-Mz-Sy-CC-C), the amount of export tax payments, gross margin, negative soil sulfur balance and phosphorus balance also contribute to the negative flow.

We can observe trade-offs and synergies between criteria. Economic criteria (return on working capital, economic risk, and gross margin) are in conflict with environmental criteria (soil carbon balance, net

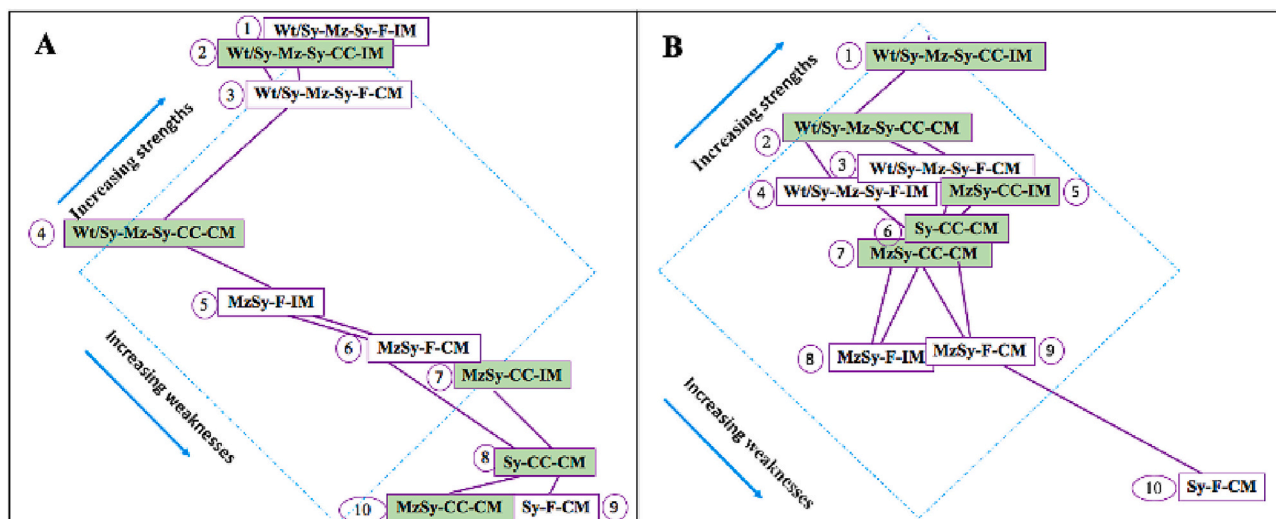


Fig. 1. Flowchart depicting cropping systems for both the production-oriented decision profile (A) and the environmental-oriented decision profile (B), incorporating the criteria values and weights. The visualization was created using the VISUAL PROMETHEE free software. Green rectangles represent cover crop cropping systems, while white rectangles represent non-cover crop cropping systems. The light blue dotted square represents the positive flow, while the negative flow of each cropping system is depicted by another square. Positive flow scores increase from the left to the top corner (indicating increasing strengths), while negative flow scores increase from the left to the bottom corner (representing increasing weaknesses). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Note: Sy: Soybean; MzSy: Maize-Soybean; Wt/SyMzSy: Wheat/Soybean-Maize-Soybean; F: Fallow; CC: Cover crops; IM: Intensified Management; CM: Conventional Management.

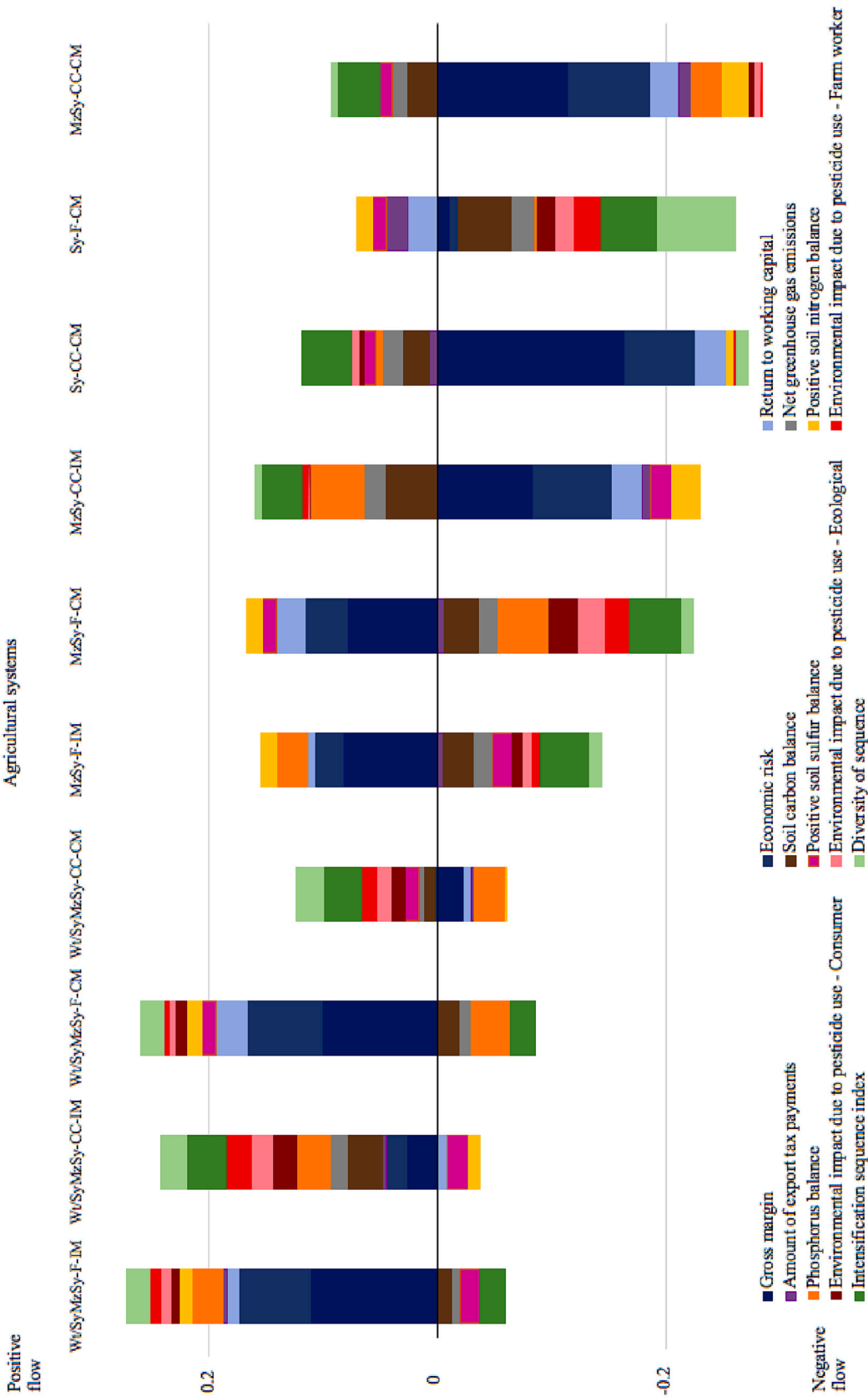


Fig. 2. Incidence of indicators assessed as positive or negative flows for each cropping system in a production-oriented decision profile. Note: Sy: Soybean; MzSy: Maize-Soybean; Wt/SyMzSy: Wheat/Soybean-Maize-Soybean; F: Fallow; CC: Cover crops; I: Intensified; C: Conventional. Positive and negative outperforming flows are shown above and below the zero line, respectively. The size of the bars is proportional to the positive and negative outperforming flows of each criterion to the performance.

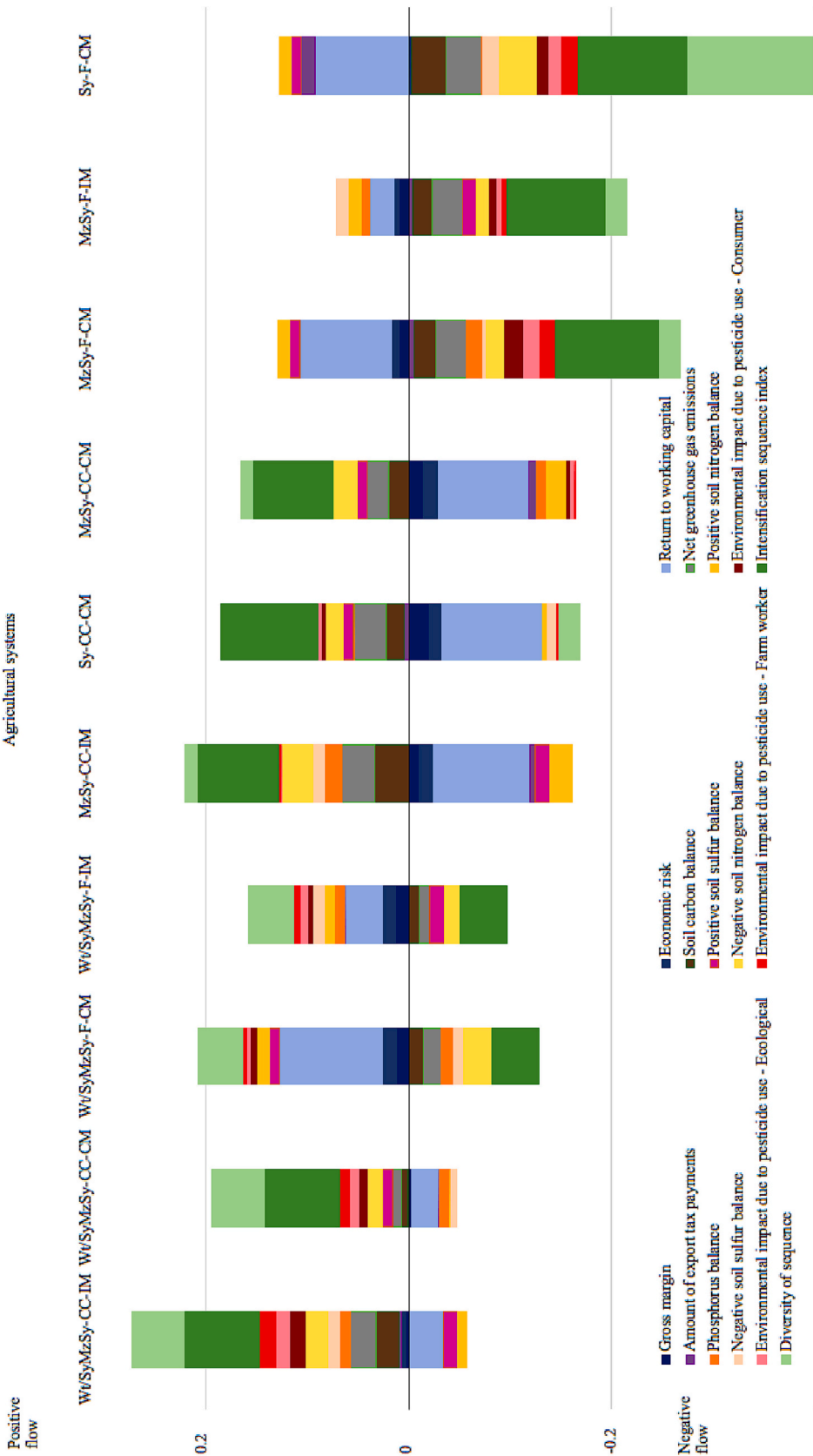


Fig. 3. Incidence of indicators assessed as positive or negative flows for each cropping system in an environmentally-oriented decision profile. Note: Sy: Soybean; MzSy: Maize-Soybean; Wt/SyMzSy: Wheat/Soybean-Maize-Soybean; F: Fallow; CC: Cover crops; I: Intensified; C: Conventional. Positive and negative outperforming flows are shown above and below the zero line, respectively. The size of the bars is proportional to the positive and negative outperforming flows of each criterion to the performance.

greenhouse gas emissions, and intensification sequence index). Cropping systems with winter fallows favor economic criteria, whereas those with cover crops favor environmental criteria (See Annexes C and D).

4. Discussion

The most diverse crop sequence (Wt/Sy-Mz-Sy) proved to be superior in a comprehensive analysis, even though it did not outperform the rest of the crop sequences in all the criteria within a framework of sustainable intensification assessment. Regarding the inclusion of cover crops, we found a trade-off between environmental and economic criteria. Intensified management outperformed conventional management in ten indicators but was inferior in the remaining ones. The PROMETHEE tool enabled the operationalization of the SI concept by ranking cropping systems based on criteria and their perceived importance. Therefore, PROMETHEE proved to be a valuable tool for group discussions on the assessment of various systems or productive scenarios

4.1. Performance of crop sequences

The Wt/Sy-Mz-Sy crop sequence demonstrated superior performance in most of the evaluated criteria. This advantage was consolidated both in the objective information (criteria) (Table 3) and in its combination with the subjective information (criteria weighting) (Fig. 2).

The Wt/Sy-Mz-Sy sequence outperformed the Mz-Sy and Sy-Sy sequences in terms of economic criteria. This is mainly due to the incorporation of an additional cash crop in the same growing season. Instead of growing only one crop in the warm season, double cropping allows growing an additional crop during the cool season, in this case wheat (Wt/Sy). Hence, double cropping helps to increase the total gross margin. In addition, the inclusion of the Wt/Sy double crop in the sequence, – rather than growing only Sy, for instance – provides greater productivity stability. This is due to the fact that the total yields are defined at two different stages of the growing season, reducing the economic risk (Table 3).

In the socioeconomic criterion amount of export tax payments, Sy-Sy ranked first due to the fact that soybean crops are taxed at the highest percentage (Annex A) and are entirely exported. Argentina is among the few countries, alongside Ghana and India, where taxes are applied to agricultural production. In contrast, most countries around the world provide financial support and subsidies to sustain agriculture (FAO, et al., 2021).

In terms of environmental criteria, the Wt/Sy-Mz-Sy crop sequence exhibited, on average, lower net greenhouse gas emissions than the Mz-Sy and Sy-Sy sequences. Although the Wt/Sy-Mz-Sy sequence ranked second regarding soil carbon balance, differing only slightly with Mz-Sy, it had lower N₂O emissions from fertilization, mineralization and residues, which resulted in lower net greenhouse gas emissions in the Wt/Sy-Mz-Sy sequence. The Sy-Sy sequence had the lowest gas emissions due to lower N fertilizer application compared to maize and wheat (Alvarez et al., 2012). However, the Sy-Sy sequence had still higher net greenhouse gas emissions due to the loss of soil carbon. Higher N₂O emissions in sequences including maize crops are consistent with other studies (Parkin and Kaspar, 2006; Gregorich et al., 2008) and differ from the results found by Piccinetti et al. (2021), where Sy-Sy and sequences that included maize and wheat showed similar cumulative N₂O emissions in absolute terms. These differences arise because the authors recalculated the emission factor for nitrogen fertilizer, as they found two weaknesses in the factor calculated by the IPCC. On one hand, they mention that the processes that occur in different crop sequences were not taken into account, such as the duration and processes during fallow periods or the inclusion of cover crops. On the other hand, they mention that biological fixation inputs to the system were not considered. Therefore, our estimation using the IPCC methodology has limitations, and in the results, we are underestimating emissions in soybean cultivation. This would indicate that the Wt/Sy-Mz-Sy sequence has even

lower net greenhouse gas emissions than the Mz-Sy and Sy-Sy sequences.

In terms of phosphorus balance, the Wt/Sy-Mz-Sy sequence tended to be neutral. This outcome can be either positive or negative depending on the soil's initial P content. However, neutral balances are desirable.

The Sy-Sy sequence exhibited lower values in terms of positive soil nitrogen balance compared to both Wt/Sy-Mz-Sy and Mz-Sy. Likewise, the Sy-Sy sequence exhibited a lower positive soil sulfur balance compared to both Mz-Sy and Wt/Sy-Mz-Sy. These findings indicate that nitrogen and sulfur inputs in fertilization were higher than exports in harvested grains, and the positive balances of these nutrients may pose a risk of leaching due to their high mobility in the soil (Carciochi et al., 2015; Portela et al., 2016). Despite the fact that the Wt/Sy-Mz-Sy sequence ranked first in both decision profiles (Figs. 2 and 3), the positive nitrogen and sulfur soil balances have negative implications, highlighting the need for adjusting fertilization doses.

The Sy-Mz-CC-IM crop sequence had lower absolute values in negative soil nitrogen and sulfur balance. On the other hand, the Sy-Sy-F-CM cropping system was found to be the most extractive in terms of nitrogen and sulfur nutrients. These findings align with those of Austin et al., 2006, which showed that although there were increased nitrogen inputs from biological fixation, the losses due to seed export in soybean resulted in a net loss of nitrogen.

The Wt/Sy-Mz-Sy sequence had, on average, a lower environmental impact than the Mz-Sy and Sy-Sy sequences. This difference was due to the lower toxicity of the pesticide package applied in the Wt/Sy-Mz-Sy sequence compared to the Mz-Sy and Sy-Sy sequences. Our results agree with previous research, which reported that the wheat/soybean double crop presents the lowest toxicity among the crop sequences grown in the study region (Principiano and Acciaresi, 2018). Ferraro et al. (2003) found that wheat had a lower environmental impact than maize and sunflower in the Pampean agroecosystem in terms of pesticide use, which is consistent with our results. The lower toxicity of the wheat/soybean double crop, and therefore of the Wt/Sy-Mz-Sy sequence, is associated with the rate of intensification. The period of land occupation is maximum in this sequence, decreasing weed abundance and the amount of herbicide applied. The Wt/Sy-Mz-Sy sequence also presented the highest diversity compared to the Mz-Sy and Sy-Sy sequences due to the incorporation of wheat.

The Wt/Sy-Mz-Sy sequence was found to be superior in both economic and environmental indicators, leading to its placement in the top four positions in both decision profiles. These findings, based on an integral assessment of cropping systems, further support the benefits of the Wt/Sy-Mz-Sy crop sequence, which is consistent with previous research approaching the topic from an ecophysiological perspective (Caviglia et al., 2004; Caviglia and Andrade, 2010; Andrade et al., 2015, 2017).

4.2. Comparison between intensified and conventional managements

In terms of rate of return, conventional management outperformed intensified management due to the use of less expensive pesticides and lower fertilization doses, leading to reduced costs and an improved rate of return.

Evaluating these systems regarding their alignment to sustainable intensification, we found in the IM treatments, highly positive soil nitrogen and sulfur balance. We consider that in future trials it would be interesting to include lower nitrogen and sulfur input rates, as IS alternative.

4.3. Cover crops analysis

Growing cover crops before warm-season crops improved environmental criteria, but worsened economic indicators. Cover crops notably improved the soil carbon balance by the amount of biomass they produced (Wander and Traina, 1996; Ding et al., 2006). Cover crop treatments, on average, had lower net greenhouse gas emissions (Abdalla et al., 2019; Restovich et al., 2019; Poeplau and Don, 2015). When

analyzing the composition of net greenhouse gas emissions, it was found that N₂O emissions from fertilization and waste, CO₂ emissions from urea, and CO₂, N₂O, and CH₄ emissions from fossil fuel use in machinery in the field were higher when cover crops were used. However, the lower N₂O emissions from the mineralization of soil organic matter and the positive soil carbon balance outweighed these emissions, resulting in lower net greenhouse gas emissions overall. In addition, cover crops contributed to a decrease in the environmental impact of cropping systems by reducing the use of pesticides.

These results are consistent with those of Principiano and Acciaresi (2017) and Baigorria et al. (2019). Additionally, these cover crops increased sequence diversity and decreased the bare soil period, thereby increasing the rate of intensification of the sequence.

The incorporation of cover crops in cropping systems led to a decrease in economic criteria such as gross margin, return on working capital, and economic risk. This can be attributed to the cost associated with cover crops, which do not provide returns in the short to mid-term, even if the yield of the following crop is maintained or decreased (Bojanich et al., 2010; Baigorria et al., 2012). Furthermore, the yield analysis of this experiment revealed that cover crops result in yield penalties, especially affecting subsequent maize crops (Hisse et al., 2022).

The adoption of cover cropping in Argentina is currently low, only 0.02% (Ministry of Agriculture, 2022). This may be due to the decrease in economic criteria, particularly gross margins, associated with this practice. Additionally, cover cropping is a relatively new practice in Argentina, and despite the willingness of Pampean farmers to incorporate new technologies, they need to adapt agronomic practices, such as planting date, drying date, species, and proportion, to make them work better for each cropping system.

In the profile prioritizing environmental factors, there is a clear preference for treatments that involve cover crops, while the productivity profile favors treatments without these crops. One solution to achieve the preferences of the environmental approach profile is to implement policies incentivizing cover crops, such as, removing the amount of export tax payments. By doing so, farmers would not have to bear the economic cost and would be more likely to adopt the practice. As the benefits become apparent and farmers gain knowledge and experience, the policy measure to incentivize cover cropping could be removed in the long run.

4.4. Further research

The mid-term experiment assessed in this study was carried out during five consecutive cropping seasons, which differed in weather conditions. In such a situation, the presence each year of all crop types in the crop sequences, provided a valuable dataset to characterize the cropping systems.

A possible improvement of the experiment could involve fine-tuning the fertilization schemes in the intensified management, especially with regard to the surpluses of nitrogen and sulfur in the soil. It would have been interesting to include the option of intensified management for soybean monocropping to assess how it ranks compared to the other systems. Furthermore, it is important to note that the long-term effects of the cropping systems should also be taken into consideration, as they may not have been detected during the assessed period.

The adoption of cover crops often incurs an economic cost for farmers. One possible solution to reduce this cost is to consider alternative options for the winter period, such as including different cash crops like field pea or lentil, instead of relying solely on legume-based cover crops.

In addition, it would be interesting to test the PROMETHEE tool by sharing the ranking results in discussion sessions with different

stakeholders, such as groups of agricultural advisors and farmers, to aid in land use planning, exploring cropping systems, and resolving conflicts related to the definition of public policies. Moreover, it would be useful to assess how political measures, such as eliminating export tax payments, could affect the ranking of cropping systems. It is important to note that the results of this study are specific to the rolling pampas, the corn belt of Argentina, but they could potentially be extrapolated to similar cropping systems and climates. The same analysis process can also be applied to other systems or climates.

It is important to acknowledge that environmental indicators for soil carbon and nutrient balances, greenhouse gas emissions, and environmental impact due to pesticide use are based on models that simplify the complex processes in agroecosystems. While real observed data or more sophisticated models could provide more accurate measurements of environmental impacts, this kind of data is often unavailable for integrated assessments. In the current study, we utilized models documented in recent research for the study region, considered reliable methods for comparing cropping systems.

5. Conclusion

In this study, it was found that more diverse crop sequences outperformed less diverse ones, particularly soybean monocropping. As a result, the Wt/Sy-Mz-Sy sequence proved to be the most aligned to the sustainable intensification trajectory for the study region. Our findings reinforce the benefits observed in previous studies and provide a more comprehensive assessment of that crop sequence. Intensified management outperformed conventional management, except in rate of return and positive sulfur and nitrogen soil balance. Although cover crop adoption can provide environmental benefits, the associated costs discourage their implementation in production-oriented decision profiles. Overall, this study is significant as it successfully utilized the concept of sustainable intensification to compare cropping systems in the rolling Pampas.

The assessment of IS plays a crucial role in the implementation of sustainable cropping systems. Multicriteria methods are useful in handling the complexity of this multi-dimensional assessment, providing valuable information for private and public decision-making. PROMETHEE is a tool that enables the identification of the strengths and weaknesses of different cropping systems, the analysis of trade-offs between criteria, and the ranking of alternative cropping systems. To effectively utilize PROMETHEE for farm decision-making and to establish consensus among stakeholders, the guidance of an expert group is of utmost importance in explaining the model building process and interpreting results.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Detail of the parameters used in the calculation of the criteria

A.1. Gross margin

Gross margin: Gross income – Direct costs (expenditure on inputs, services, harvesting, and marketing) (Ghida Daza, 2009).

Gross income is determined by multiplying the yield of each crop by the market price. The market price used is based on the price available at the time of harvest in Bolsa de Comercio de Rosario, Argentina (<https://www.bcr.com.ar/es/mercados/mercado-de-granos/cotizaciones/cotizaciones-locales-0>).

Direct costs, including inputs (see Annex C) and labor, were valued at the time of their utilization. Marketing costs were determined based on the rates provided by the Federated Farmers of Argentina (AFA: Agricultores Federados Argentinos, available at <http://diario.afascl.coop/afaw/index.vsp>). Freight costs were calculated using the rates provided by the Argentine Confederation of Automotive Loads Transport (CATAC: Confederación Argentina de Transporte Automotor de Cargas, available at <https://www.catac.org.ar/costos-y-tarifas>).

All prices are expressed in US dollars. It's important to note that the cost of land rent is not included in the direct cost calculations.

The data is available at:

https://docs.google.com/spreadsheets/d/1vXAWA9ssfRuNsbh4zmI9s7kXG55JUBr2/edit?usp=share_link&ouid=105997147179329789179&rtpof=true&sd=true

A.2. Economic Risk

The economic risk was assessed by considering the minimum annual gross margin over a 5-year period.

Gross Margin (shaded area indicates the value used for economic risk assessment).

Cropping systems	Cropping seasons				
	2015/2016	2016/2017	2017/2018	2018/2019	2019/2020
Sy-F-CM	369,7	621,9	271,8	424,8	183,9
Sy-CC-CM	145,4	232,9	79,0	229,5	30,5
MzSy-F-CM	677,8	731,7	316,5	456,4	358,5
MzSy-F-IM	556,4	882,6	278,6	523,2	328,9
MzSy-CC-CM	262,0	347,5	-0,8	341,2	143,8
MzSy-CC-IM	236,8	425,7	8,0	425,2	225,2
Wt/SyMzSy-F-CM	606,5	803,4	417,6	483,8	397,2
Wt/SyMzSy-F-IM	610,5	759,7	421,7	596,7	392,6
Wt/SyMzSy-CC-CM	339,2	523,9	237,6	464,5	213,6
Wt/SyMzSy-CC-IM	407,8	510,9	258,0	636,3	344,3

A.3. Return on Working Capital

The return on working capital is determined by dividing the gross margin by the cost of inputs and services.

A.4. Amount of Export Tax Payments

The amount of export tax payments was calculated by multiplying the gross income of each crop by the respective tax rate. For soybean production, 100% of the production was considered. However, for maize and wheat production, only 60% was taken into account, as these crops are not entirely exported due to their internal use in industries and animal feed.

Example calculation:

Cropping seasons 2015/2016, Crop Wheat.

Gross income 567,54 multiplied by 0,23 and then by 0,6. Export tax = 78,32.

Export tax rate specific to each product and varied throughout the period considered.

Crop	Cropping seasons				
	2015/2016	2016/2017	2017/2018 -%-	2018/2019	2019/2020
Soybean	30	30	28	28	33
Maize	0	0	0	10	12
Wheat	23	0	0	10	7

A.5. Soil carbon balance

We estimated the carbon balance using the AMG model, a simple model designed to simulate SOC evolution (Andriulo et al., 1999; Milesi Delaye et al., 2013).

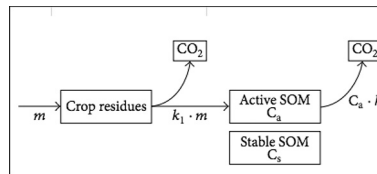


Diagram of AMG model (Andriulo et al., 1999; Milesi Delaye et al., 2013).

AMG model of the evolution of soil organic carbon (SOC): Stable carbon stock + mass of carbon contributed by the crop * humification coefficient/mineralization coefficient. (Andriulo et al., 1999; Milesi Delaye et al., 2013).

Initial carbon stock: At the beginning of the trial, stable carbon was measured in the soil organic matter at a depth of 0–20 cm using the Leco method (Leco autoanalyzer model TRU SPEC, Leco Corp., USA).

Block	Initial C (tn C/ha)
1	33,95
2	35,56
3	35,01

Mass of carbon contributed by the crop residues (both aerial and underground) was calculated annually.

Mass crop residues (tn C/ha): Root biomass + Aerial biomass

Root biomass: Dry matter yield / Harvest index * 0.3 (a root biomass/aerial biomass ratio of 30% for all crops considered, which includes rhizodepositions (Bolinder et al., 2007)) * 0.4 (C content of 40% of plant biomass).

Aerial biomass: (Dry matter yield / Harvest index - Dry matter yield) * 0.4 (C content of 40% of plant biomass).

The dry yield and harvest index were determined by conducting measurements during the trial.

The data is available at:

https://docs.google.com/spreadsheets/d/1vXAWA9ssfRuNsbh4zmI9s7kXG55JUBr2/edit?usp=share_link&ouid=105997147179329789179&rtpof=true&sd=true

Humification coefficient in direct sowing (k1)

Crop	k1	Fuente
Maize	0,13	Mary and Wylleman, 2001
Wheat	0,13	Milesi Delaye et al., 2013
Soybean	0,17	Milesi Delaye et al., 2013
Soybean 2nd	0,17	Milesi Delaye et al., 2013
Cover crop	0,25	consulted to Luis Milesi 4/29/2021

Mineralization coefficient (k) in direct sowing = 0,078 (Milesi Delaye et al., 2013)

Stable carbon: Initial carbon – (Initial carbon * 0,25)

0.25 because MOS active 11,1 tn C/ha (25% of SOC) (Milesi Delaye et al., 2013; Andriulo et al., 2012)

Carbon Balance: Input – Lost.

Input: Mass of carbon contributed by the crop * k1.

Lost: k * (Initial Carbon - Stable carbon).

Carbon Balance

Cropping systems	Cropping seasons				
	15–16	16–17	17–18	18–19	19–20
Sy-F-CM	–0,35	–0,15	–0,30	–0,31	–0,39
Sy-CC-CM	0,34	0,21	0,45	0,29	0,42
MzSy-F-CM	–0,16	0,00	–0,25	–0,09	–0,26
MzSy-F-IM	–0,17	0,08	–0,20	–0,07	–0,21
MzSy-CC-CM	0,33	0,21	0,39	0,34	0,62
MzSy-CC-IM	0,29	0,32	0,61	0,52	0,99
Wt/SyMzSy-F-CM	–0,05	0,15	–0,13	–0,09	–0,11
Wt/SyMzSy-F-IM	0,09	0,14	–0,06	–0,08	–0,10
Wt/SyMzSy-CC-CM	0,20	0,18	0,21	0,11	0,39
Wt/SyMzSy-CC-IM	0,28	0,29	0,35	0,32	0,81

A.6. Net greenhouse gas emissions

Net emissions (Carbon equivalent): Carbon balance + N₂O emissions from N fertilization + N₂O emissions from mineralization + CO₂ emissions from urea fertilization + CO₂, N₂O and CH₄ emissions from fossil fuel use in machinery passes in the field + N₂O emissions from stubble input. 2006 and 2019 IPCC guidelines (Eggleston et al., 2006, Shukla et al., 2019).

Carbon balance (Carbon equivalent): Carbon balance (see annex A.5) * -3,6 (Volume 4 Chapter 2 IPCC 2006 and 2019)

N₂O emissions from N fertilization (Carbon equivalent): N fertilization (see annex E) * 0.016 (Table 11.1 Volume 4 Chapter 11 IPCC 2019) * 44/28 * 298 (Heating power CO₂ eq)

N₂O emissions from mineralization (Carbon equivalent): (Negative carbon balances * 1/10) * 1000) * 0,006 (Table 11.1 Volume 4 Chapter 11 IPCC 2019) * 44/28 * 298 (Heating power CO₂ eq)

CO₂ emissions from urea fertilization: tn urea (see annex E) * 0,2 (Table 3.31 Volume 4 Chapter 11 IPCC 2006) * 44/12 (to convert CO₂-C emissions into CO₂).

CO₂, N₂O and CH₄ emissions from fossil fuel use in machinery passes in the field: Machinery passes * Diesel oil consumption according to agricultural work * 0,0000402 (Conversion factor (TJ/unit)).

CO₂ kg/TJ: Result * 74100/1.2534

CH₄ kg/TJ: Result * 4,15/1.2534

N₂O kg/TJ: Result * 28,6 /1.2534

(Table 3.31 Volume 2 Chapter 3 IPCC 2006)

CO₂ eq: CO₂ kg/TJ + CH₄ kg/TJ /25 + N₂O kg/TJ/ 298 (Heating power CO₂ eq)

N₂O emissions from stubble input: (N biomass – N exported in grain) * 0,006 (Table 11.1 Volume 4 Chapter 11 IPCC 2019) * 44/28 * 298 (Heating power CO₂ eq)

N exported in grain (kg N ha⁻¹): Dry yield (kg ha⁻¹) * N in grain (gr/kg) /1000

N biomass: N exported in grain (kg N ha⁻¹) / % N in plant

The dry yield, N in grain and % N in plant were obtained by measuring them in the trial

A.7. Phosphorus balance

Phosphorus balance: P fertilizers input (see Annex E) - P output by harvested grain

Harvested grain: Dry yield * P in grain

The dry yield and phosphorus content in the grain were determined by conducting measurements during the trial.

The data is available at:

https://docs.google.com/spreadsheets/d/1vXAWA9ssfRuNsbh4zmI9s7kXG55JUBr2/edit?usp=share_link&ouid=105997147179329789179&rtpof=true&sd=true

A.8. Sulfur balance

Sulfur balance: S fertilizers input (see Annex E) + Atmospheric deposition (Carnelos et al., 2019) - S output by harvested grain.

Harvested grain: Dry yield * S in grain.

The dry yield and sulfur content in the grain were determined by conducting measurements during the trial.

The data is available at:

https://docs.google.com/spreadsheets/d/1vXAWA9ssfRuNsbh4zmI9s7kXG55JUBr2/edit?usp=share_link&ouid=105997147179329789179&rtpof=true&sd=true

A.9. Nitrogen balance

Nitrogen balance: N fertilizers input (see Annex E) + Atmospheric deposition (Carnelos et al., 2019) + Biological Fixation (Soybean crop Collino et al., 2015; Vicia villosa cover crop, Enrico et al., 2020) - N output by harvested grain – Volatilization (Alvarez and Steinbach, 2006; Ferraris et al., 2014)

Harvested grain: Dry yield * N in grain

The dry yield and N in grain were obtained by measuring them in the trial

The data is available at:

https://docs.google.com/spreadsheets/d/1vXAWA9ssfRuNsbh4zmI9s7kXG55JUBr2/edit?usp=share_link&ouid=105997147179329789179&rtpof=true&sd=true

A.10. Environmental impact due to pesticide use

Based on the agrochemicals used (see details Annex F) the EIQ was calculator available on: <https://cals.cornell.edu/new-york-state-integrate-d-pest-management/risk-assessment/eiq/eiq-calculator>.

EIQ = Farm Worker {C* ((DT*5) + (DT * P))} + Consumer {(C* ((S + P)/2) *SY) + L} + Ecology risk {(F*R) + (D* ((S + P)/2) *3) + (Z*P*3) + (B*P*5)} / 3

DT = dermal toxicity, C = chronic toxicity, SY = systemicity, F = toxicity to

fish, L = leaching potential, R = surface loss potential, D = toxicity to

birds, S = half-life in soil, Z = toxicity to bees, B = toxicity to

beneficial arthropods, and P = half-life of the surface of the plant

In the multicriteria analysis, the risk toxicities for workers, consumers, and the environment were included separately.

Appendix B. Sensitivity analysis

As suggested in the PROMETHEE manual, Stability Intervals were utilized to conduct the sensitivity analysis. These intervals allow for assessing the impact of weight changes on the analysis (PROMETHEE, 2013). The tables present the assigned weights for each criterion and the range within which they can be modified without altering the order of the cropping systems. For instance, in the decision profile with a production-oriented approach, the gross margin is assigned a weight of 25%. The cropping system that ranks first will remain unchanged as long as the weight varies from 23 to 100. This demonstrates that minor adjustments in the weight of each criterion do not affect the order of the cropping systems.

B.1. Stability range of weights for the production-oriented decision profile

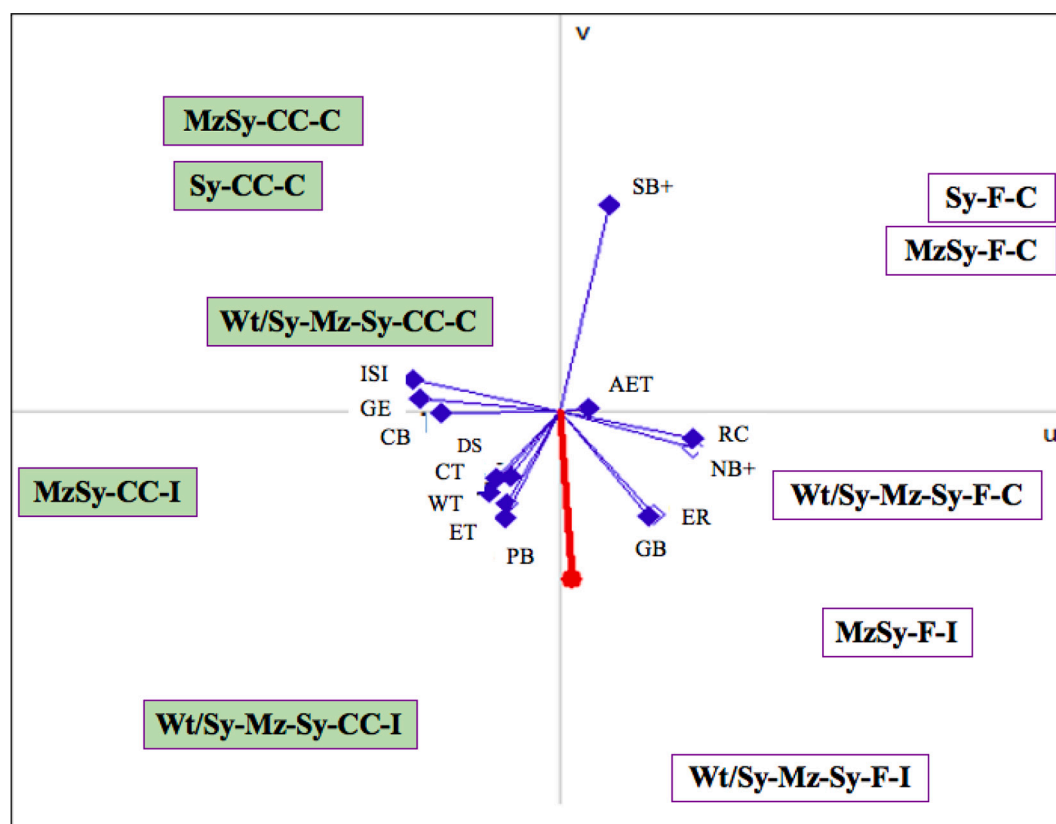
Criterion	Assigned weight	Weight change range within which the ranking of the subsequent remains unchanged									
		1st	1st & 2nd	1st to 3rd	1st to 4	1st to 5	1st to 6	1st to 7	1st to 8	1st to 9	1st to 10
Criterion						–%–					
Gross margin	25	23–100	23–32	23–32	23–32	23–32	23–32	23–32	23–29	24–29	24–29
Economic risk	12	11–78	11–19	11–19	11–19	11–19	11–19	11–19	11–19	12–19	12–19
Return on working capital	5	3–14	3–9	3–9	3–9	3–9	4–9	4–9	4–8	5–8	5–8
Amount of export tax payments	3	0–29	0–29	0–29	0–27	0–20	0–15	0–13	0–10	3–10	3–10
Soil carbon balance	8	0–9	3–9	3–9	3–9	3–9	3–9	3–9	4–9	4–8	4–8
Net greenhouse gas emissions	4	0–5	0–5	0–5	0–5	0–5	0–5	0–5	0–5	0–4	0–4
Soil phosphorus balance	8	3–24	4–24	4–24	4–15	4–15	4–9	4–9	4–9	8–9	8–9
Positive soil sulfur balance	4	2–33	2–8	2–8	2–8	2–8	2–8	2–8	2–8	3–8	3–8
Positive soil nitrogen balance	3	0–6	0–6	0–6	0–6	0–6	2–6	2–6	2–6	2–6	1–5
Environmental impact due to pesticide use - Consumer	4	0–6	0–6	0–6	0–6	0–6	0–6	0–6	0–6	0–5	0–5
Environmental impact due to pesticide use - Ecological	4	0–7	0–7	0–7	0–7	0–7	0–6	0–6	0–6	0–5	0–5
Environmental impact due to pesticide use - Farm worker	4	0–6	0–6	0–6	0–6	0–6	0–6	0–6	0–6	0–4	0–4
Intensification sequence index	8	0–9	4–9	4–9	4–9	4–9	4–9	4–9	5–9	5–9	5–8
Diversity	8	0–22	0–22	0–22	0–22	0–22	0–15	0–15	3–15	3–8	3–8

B.2. Stability range of weights for the environmental e-oriented decision profile

Criterion	Assigned weight	Weight change range within which the ranking of the subsequent positions remains unchanged									
		1st	1st & 2nd	1st to 3rd	1st to 4	1st to 5	1st to 6	1st to 7	1st to 8	1st to 9	1st to 10
Criterion						–%–					
Gross margin	3	0–33	0–14	0–14	3–14	3–14	3–10	3–10	3–10	3–10	3–9
Economic risk	3	0–28	0–14	1–14	2–14	2–14	2–14	2–14	2–14	2–14	2–14
Return on working capital	18	0–31	0–25	16–25	18–25	18–25	18–25	18–25	18–25	18–25	18–25
Amount of export tax payments	2	0–49	0–29	0–13	1–13	1–9	1–9	1–9	1–9	1–9	1–9
Soil carbon balance	6	0–50	0–22	0–8	0–6	0–6	0–6	0–6	0–6	0–6	0–6
Net greenhouse gas emissions	6	0–58	0–22	0–8	0–6	0–6	0–6	0–6	0–6	0–6	0–6
Soil phosphorus balance	3	0–45	0–12	0–5	0–4	0–4	0–4	0–4	0–4	0–4	0–4
Positive soil nitrogen balance	3	0–19	0–15	1–15	3–15	3–15	3–11	3–11	3–10	3–11	3–10
Negative soil nitrogen balance	6	0–60	0–32	0–8	0–6	0–6	0–6	0–6	0–6	0–6	0–6
Positive soil sulfur balance	2	0–8	0–8	0–8	0–6	0–6	0–6	0–6	2–6	2–6	2–6
Negative soil sulfur balance	3	0–100	0–13	0–5	0–5	0–5	0–5	0–5	0–3	0–3	0–3
Environmental impact due to pesticide use - Consumer	3	0–100	0–100	0–100	2–100	2–35	2–35	2–35	2–4	2–4	2–4
Environmental impact due to pesticide use - Farm worker	3	0–100	0–100	0–21	1–21	2–21	1–21	1–21	1–5	1–5	1–5
Environmental impact due to pesticide use - Ecological	3	0–100	0–100	0–15	2–15	2–15	2–15	2–15	2–4	2–4	2–4
Diversity	18	0–100	0–100	7–100	17–100	17–100	17–24	17–24	17–24	17–24	17–24
Intensification sequence index	18	0–71	9–62	9–20	9–18	9–18	9–18	9–18	9–18	9–18	9–18

Appendix C. GAIA Plan (Geometrical Analysis for Interactive Aid), for the production-oriented decision profile considering the values of each criterion and their weighting

The result was generated using the VISUAL PROMETHEE software. The decision axis, depicted as the red vector, represents the weighting of the criteria. Its direction indicates the most favorable cropping systems, as identified by the criteria proposed by Mareschal and De Smet (2009).

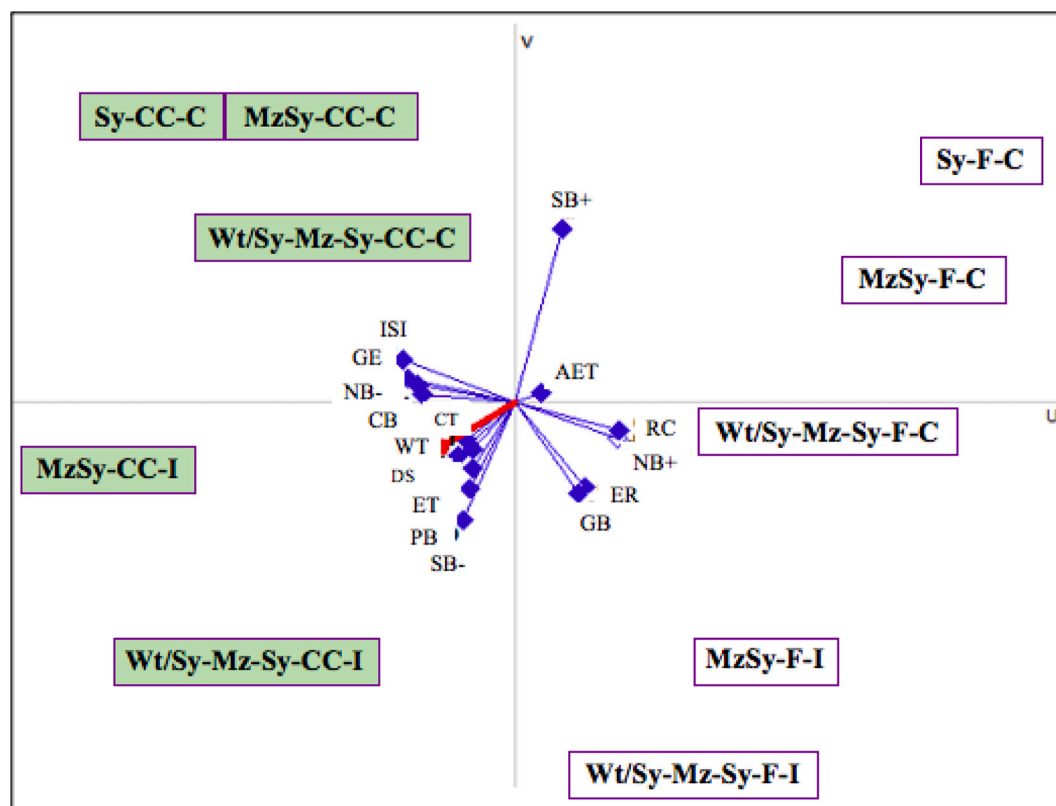


Note: **SB+** Positive soil sulfur balance, **AET** Amount of export tax payments, **RC** Return on working capital, **NB+** Positive soil nitrogen balance, **ER** Economic risk, **GB** Gross margin, **PB** Phosphorus balance, **ET** Ecological toxicity, **WT** Worker toxicity, **CT** Consumer toxicity, **DS** Diversity of sequence, **CB** Soil carbon balance, **GE** Net greenhouse gas emissions, **ISI** Intensification sequence index. **U** is the first principal component, it contains the maximum possible quantity of information, and **V** is the second principal component, providing the maximum additional information orthogonal to **U**.

Sy: Soybean; MzSy: Maize-Soybean; Wt/SyMzSy: Wheat/Soybean-Maize-Soybean; F: Fallow; CC: Cover crops; IM: Intensified management; CM: Conventional management.

Appendix D. GAIA Plan (Geometrical Analysis for Interactive Aid), for the environmentally-oriented decision profile considering the values of each criterion and their weighting

The result was generated using the VISUAL PROMETHEE software. The decision axis, depicted as the red vector, represents the weighting of the criteria. Its direction indicates the most favorable cropping systems, as identified by the criteria proposed by [Mareschal and De Smet \(2009\)](#).



Note: **SB+** Positive sulfur soil balance, **AET** Amount of export tax payments, **RC** Return on working capital, **NB+** Positive nitrogen soil balance, **ER** Economic risk, **GB** Gross margin, **SB-** Negative sulfur balance, **PB** Phosphorus balance, **ET** Ecological toxicity, **WT** Worker toxicity, **CT** Consumer toxicity, **DS** Diversity of sequence, **CB** Soil carbon balance, **NB-** Negative nitrogen balance, **GE** Net greenhouse gas emissions, **ISI** Intensification sequence index. **U** is the first principal component, it contains the maximum possible quantity of information, and **V** is the second principal component, providing the maximum additional information orthogonal to **U**.

Sy: Soybean; MzSy: Maize-Soybean; Wt/SyMzSy: Wheat/Soybean-Maize-Soybean; F: Fallow; CC: Cover crops; IM: Intensified management; CM: Conventional management.

Appendix E. Fertilization schemes for evaluated crops grown under two management systems (CM: conventional management, IM: intensified management) from 2015/16 to 2019/20

Year	System	Soybean	Maize	Wheat	Soybean DC ¹
2015/16	CM	16 kg P ha ⁻¹ as TSP, at sowing.	69 kg N ha ⁻¹ as urea, at sowing, and 36.8 kg N ha ⁻¹ as urea, at V6; 16 kg P ha ⁻¹ as TSP, at sowing.	69 kg N ha ⁻¹ as urea, at early tillering; 16 kg P ha ⁻¹ as TSP, at sowing.	–
	IM	34.1 kg P ha ⁻¹ as TSP, at sowing; 15.2 kg S ha ⁻¹ as GG, previous sowing.	69 kg N ha ⁻¹ as urea, at sowing, and 36.8 kg N ha ⁻¹ as urea, at V6; 16 kg P ha ⁻¹ as SSP, at sowing; 15.2 kg S ha ⁻¹ as GG, previous sowing.	115 kg N ha ⁻¹ as urea, at early tillering; 34 kg P ha ⁻¹ as TSP, at sowing; 17.1 kg S ha ⁻¹ as GG, at sowing.	–
2016/17	CM	16 kg P ha ⁻¹ as TSP, at sowing.	69 kg N ha ⁻¹ as urea, at sowing, and 36.8 kg N ha ⁻¹ as urea, at V6.	69 kg N ha ⁻¹ as urea, at early tillering; 18 kg P ha ⁻¹ as TSP, at sowing.	–
	IM	34 kg P ha ⁻¹ as TSP, at sowing; 15.2 kg S ha ⁻¹ as GG, previous sowing.	115 kg N ha ⁻¹ as urea, at sowing, and 46 kg N ha ⁻¹ as urea, at V6; 16 kg P ha ⁻¹ as SSP, at sowing; 37.1 kg S ha ⁻¹ as SSP and GG, previous sowing.	115 kg N ha ⁻¹ as urea, at early tillering; 34 kg P ha ⁻¹ as TSP, at sowing; 17.1 kg S ha ⁻¹ as GG, at sowing.	–
2017/18	CM	16 kg P ha ⁻¹ as TSP, at sowing.	69 kg N ha ⁻¹ as urea, at sowing, and 36.8 kg N ha ⁻¹ as urea, at V6.	69 kg N ha ⁻¹ as urea, at early tillering; 16 kg P ha ⁻¹ as TSP, at sowing.	16 kg P ha ⁻¹ as TSP, at sowing.
	IM	19.6 kg P ha ⁻¹ and 12 kg S ha ⁻¹ as CCF, at sowing.	61.5 kg N ha ⁻¹ , at sowing, and 67.5 kg N ha ⁻¹ , at V6; 28 kg P ha ⁻¹ and 16 kg S ha ⁻¹ as CCF, previous sowing.	21 kg N ha ⁻¹ , at sowing, and 40.6 kg N ha ⁻¹ , at early tillering; 19.6 kg P ha ⁻¹ and 12 kg S ha ⁻¹ as CCF, at sowing.	8.3 kg P ha ⁻¹ and 4.8 kg S ha ⁻¹ as CCF, at sowing.
2018/19	CM	16 kg P ha ⁻¹ as TSP, at sowing.	46 kg N ha ⁻¹ as urea, at sowing, and 36.8 kg N ha ⁻¹ as urea, at V6; 16 kg P ha ⁻¹ as TSP, at sowing.	69 kg N ha ⁻¹ as urea, at early tillering; 16 kg P ha ⁻¹ as TSP, at sowing.	16 kg P ha ⁻¹ as TSP, at sowing.

(continued on next page)

(continued)

Year	System	Soybean	Maize	Wheat	Soybean DC ¹
2019/20	IM	19.6 kg P ha ⁻¹ and 12 kg S ha ⁻¹ as CCF, at sowing.	61.5 kg N ha ⁻¹ , at sowing, and 86.4 kg N ha ⁻¹ , at V6; 28 kg P ha ⁻¹ and 16 kg S ha ⁻¹ as CCF, previous sowing.	21 kg N ha ⁻¹ , at sowing, and 67.5 kg N ha ⁻¹ , at early tillering; 19.6 kg P ha ⁻¹ and 12 kg S ha ⁻¹ as CCF, at sowing	8.3 kg P ha ⁻¹ and 4.8 kg S ha ⁻¹ as CCF, at sowing.
	CM	16 kg P ha ⁻¹ as TSP, at sowing.	46 kg N ha ⁻¹ as urea, at sowing, and 36.8 kg N ha ⁻¹ as urea, at V6; 16 kg P ha ⁻¹ as TSP, at sowing.	69 kg N ha ⁻¹ as urea, at early tillering; 16 kg P ha ⁻¹ as TSP, at sowing.	16 kg P ha ⁻¹ as TSP, at sowing.
	IM	19.6 kg P ha ⁻¹ and 12 kg S ha ⁻¹ as CCF, at sowing.	61.5 kg N ha ⁻¹ , at sowing, and 67.5 kg N ha ⁻¹ , at V6; 26 kg P ha ⁻¹ and 16 kg S ha ⁻¹ as CCF, previous sowing.	21 kg N ha ⁻¹ , at sowing, and 67.5 kg N ha ⁻¹ , at early tillering; 19.6 kg P ha ⁻¹ and 12 kg S ha ⁻¹ as CCF, at sowing.	10.4 kg P ha ⁻¹ and 6.4 kg S ha ⁻¹ CCF, at sowing.

¹ CCF: complex compound fertilizer; DC: double-cropped; GG: granular gypsum; SSP: single superphosphate; TSP: triple superphosphate; V6: six-ligulated leaf stage (Ritchie et al., 1992).

Appendix F. Chemical weed management for evaluated crops grown during 2015/16–2019/20. Values in parenthesis represent dosages under cover crop (left) and fallow (right) treatments. For cover crops, values in parenthesis represent dosages when cover crops preceded maize (M) and soybean (S)

Year	Soybean	Maize	Wheat	Soybean DC ¹	Cover crops
2015/16	2,4-D ² (0; 3.5 l ha ⁻¹)	2,4-D (0; 2.8 l ha ⁻¹)	dicamba (0.12; 0.12 l ha ⁻¹)	glyphosate (4; 4 kg ha ⁻¹)	2,4-D (M: 0.4; S: 2.5 l ha ⁻¹)
	acetochlor (0; 2 l ha ⁻¹)	acetochlor (4; 4 l ha ⁻¹)	glyphosate (2.5; 2.5 l ha ⁻¹)		chlorimuron ethyl (M: 0; S: 0.20 kg ha ⁻¹)
2016/17	glyphosate (0; 9 kg ha ⁻¹)	atrazine (2; 3 kg ha ⁻¹)	glyphosate (4; 4 kg ha ⁻¹)		glyphosate (M: 9; S: 21 l ha ⁻¹)
	glyphosate (0; 16 l ha ⁻¹)	glyphosate (0; 9 l ha ⁻¹)	metsulfuron-methyl (6; 6 g ha ⁻¹)		
2017/18	metsulfuron-methyl (0; 12.5 g ha ⁻¹)	2,4-D (0; 4.7 l ha ⁻¹)	dicamba (0.12; 0.12 l ha ⁻¹)	glyphosate (2; 2 kg ha ⁻¹)	2,4-D (M: 3.2; S: 5.7 l ha ⁻¹)
	2,4-D (0; 4.7 l ha ⁻¹)	acetochlor (0; 3 l ha ⁻¹)	glyphosate (2.5; 2.5 l ha ⁻¹)		chlorimuron ethyl (M: 0; S: 0.20 kg ha ⁻¹)
2018/19	acetochlor (0; 3 l ha ⁻¹)	atrazine (2; 5 kg ha ⁻¹)	metsulfuron-methyl (8; 8 g ha ⁻¹)		glyphosate (M: 8.6; S: 11.7 l ha ⁻¹)
	glyphosate (0; 10 kg ha ⁻¹)	glyphosate (4; 4 kg ha ⁻¹)			
2019/20	glyphosate (0; 9 l ha ⁻¹)	glyphosate (4; 13 l ha ⁻¹)	2,4-D (0.8; 0.8 l ha ⁻¹)	glyphosate (4.3; 4.3 kg ha ⁻¹)	2,4-D (M: 4.6; S: 5 l ha ⁻¹)
	imazethapyr (0; 1.6 l ha ⁻¹)	2,4-D (0; 4.6 l ha ⁻¹)	dicamba (0.15; 0.15 l ha ⁻¹)	haloxyfop-R-methyl (0.15; 0.15 l ha ⁻¹)	acetochlor (M: 0; S: 3 l ha ⁻¹)
2015/16	2,4-D (0; 5.2 l ha ⁻¹)	acetochlor (5; 5 l ha ⁻¹)	glyphosate (2; 2 kg ha ⁻¹)		chlorimuron ethyl (M: 0; S: 0.20 kg ha ⁻¹)
	acetochlor (0; 3 l ha ⁻¹)	atrazine (2; 2 kg ha ⁻¹)	metsulfuron-methyl (6; 6 g ha ⁻¹)		glyphosate (M: 8.8; S: 15 kg ha ⁻¹)
2016/17	clethodim (1.5; 1.5 l ha ⁻¹)	glyphosate (7.2; 15.6 kg ha ⁻¹)			
	glyphosate (15; 30 kg ha ⁻¹)				
2017/18	haloxyfop-R-methyl (0.6; 0.6 l ha ⁻¹)	2,4-D (0; 4.6 l ha ⁻¹)	2,4-D (0.8; 0.8 l ha ⁻¹)	clethodim (1.5; 1.5 l ha ⁻¹)	2,4-D (M: 3; S: 2.3 l ha ⁻¹)
	metsulfuron-methyl (0; 9 g ha ⁻¹)	acetochlor (4; 4 l ha ⁻¹)	dicamba (0.15; 0.15 l ha ⁻¹)	glyphosate (2.8; 2.8 kg ha ⁻¹)	acetochlor (M: 0; S: 3 l ha ⁻¹)
2018/19	2,4-D (0; 5.3 l ha ⁻¹)	atrazine (2; 2 kg ha ⁻¹)	glyphosate (3.7; 3.7 kg ha ⁻¹)		chlorimuron ethyl (M: 0; S: 0.07 kg ha ⁻¹)
	acetochlor (0; 1.5 l ha ⁻¹)	glyphosate (7.4; 18.2 kg ha ⁻¹)	metsulfuron-methyl (6; 6 g ha ⁻¹)		glyphosate (M: 6.4; S: 8.7 kg ha ⁻¹)
2019/20	chlorimuron ethyl (0; 0.10 kg ha ⁻¹)				
	clethodim (2.1; 2.1 l ha ⁻¹)				
2015/16	glyphosate (15; 22.4 kg ha ⁻¹)	2,4-D (0; 3.2 l ha ⁻¹)	2,4-D (0.8; 0.8 l ha ⁻¹)		2,4-D (M: 2.2; S: 3.4 l ha ⁻¹)
	haloxyfop-R-methyl (0.6; 0.6 l ha ⁻¹)	acetochlor (3; 3 l ha ⁻¹)	dicamba (0.12; 0.12 l ha ⁻¹)		acetochlor (M: 0; S: 3 l ha ⁻¹)
2016/17	metsulfuron-methyl (0; 15 g ha ⁻¹)	atrazine (2; 2 kg ha ⁻¹)	glyphosate (1.3; 1 kg ha ⁻¹)		glyphosate (M: 7.3; S: 7.7 kg ha ⁻¹)
	2,4-D (1.6; 4 l ha ⁻¹)	glyphosate (5.2; 10.7 kg ha ⁻¹)	metsulfuron-methyl (5; 5 g ha ⁻¹)		
2017/18	acetochlor (0; 2.6 l ha ⁻¹)				
	chlorimuron ethyl (0; 0.20 kg ha ⁻¹)				
2018/19	clethodim (1.5; 1.5 l ha ⁻¹)				
	glyphosate (4.5; 14 kg ha ⁻¹)				
2019/20	metsulfuron-methyl (0; 9 g ha ⁻¹)				

¹ DC: double cropped.

² 2,4-D: (2,4-Dichlorophenoxy) acetic acid; acetochlor: 2-Chloro-N-(ethoxymethyl)-N-(2-ethyl-6-methylphenyl)acetamide; atrazine: 6-Chloro-N2-ethyl-N4-(propan-2-yl)-1,3,5-triazine-2,4-diamine; chlorimuron ethyl: ethyl 2-[(4-chloro-6-methoxy-pyrimidin-2-yl)carbamoylsulfamoyl]benzoate; clethodim: 2-[(E)-N-[(E)-3-chloro-prop-2-en-2-yl]-C-ethylcarbonimidoyl]-5-(2-ethylsulfanylpropyl)-3-hydroxycyclohex-2-en-1-one; dicamba: 3,6-Dichloro-2-methoxybenzoic acid; glyphosate: N-(phosphonomethyl)glycine; haloxyfop-R-methyl: methyl (2R)-2-(4-[(3-chloro-5-(trifluoromethyl)pyridin-2-yl)oxy]phenoxy)propanoate; imazethapyr: 5-ethyl-2-(4-methyl-5-oxo-4-propan-2-yl-1H-imidazol-2-yl)pyridine-3-carboxylic acid; metsulfuron-methyl: methyl 2-[(4-methoxy-6-methyl-1,3,5-triazin-2-yl)carbamoylsulfamoyl]benzoate.

References

- Abdalla, M., Hastings, A., Cheng, K., Yue, Q., Chadwick, D., Espenberg, M., Espenberg, M., Truu, J., Rees, R.M., Smith, P., 2019. A critical review of the impacts of cover crops on nitrogen leaching, net greenhouse gas balance and crop productivity. *Glob. Chang. Biol.* <https://doi.org/10.1111/gcb.14644>.
- Alvarez, R., Steinbach, H., 2006. Ammonia volatilization from nitrogen fertilizers in pampean agroecosystems of Argentina. In: 18th World Congress of Soil Science.
- Alvarez, C., Costantini, A., Alvarez, C.R., Alves, B.J.R., Jantalia, C.P., Martellotto, E.E., Urquiga, S., 2012. Soil nitrous oxide emissions under different management practices in the semiarid region of the Argentinian Pampas. *Nutr. Cycl. Agroecosyst.* 94 (2–3) <https://doi.org/10.1007/s10705-012-9534-9>.
- Alvarez, R., Steinbach, H.S., De Paepe, L., J., 2014. A regional audit of nitrogen fluxes in pampean agroecosystems. *Agric. Ecosyst. Environ.* 184 <https://doi.org/10.1016/j.agee.2013.11.003>.
- Andrade, J.F., Poggio, S.L., Ermácora, M., Satorre, E.H., 2015. Productivity and resource use in intensified cropping systems in the rolling pampa, Argentina. *Eur. J. Agron.* 67 <https://doi.org/10.1016/j.eja.2015.03.001>.
- Andrade, J.F., Poggio, S.L., Ermácora, M., Satorre, E.H., 2017. Land use intensification in the rolling Pampa, Argentina: diversifying crop sequences to increase yields and resource use. *Eur. J. Agron.* 82 <https://doi.org/10.1016/j.eja.2016.09.013>.
- Andriulo, A., Mary, B., Guerif, J., 1999. Modelling soil carbon dynamics with various cropping sequences on the rolling pampas. *Agronomie* 19 (5). <https://doi.org/10.1051/agro:19990504>.
- Andriulo, A., Galantini, J., Studdert, G., Sasal, M., Wilson, M., Basanta, M., Sanchez, M. C., Gudelj, V., Irizar, A., Restovich, S., 2012. Existencias de carbono orgánico edáfico bajo diferentes usos y tipos de suelo. In: *Proceedings of the XIX Congreso Latinoamericano de la Ciencia del suelo and XXIII Congreso Argentino de la Ciencia del Suelo*.
- Austin, A.T., Piñeiro, G., Gonzalez-Polo, M., 2006. More is less: Agricultural impacts on the N cycle in Argentina. In: *Biogeochemistry*, vol. 79. <https://doi.org/10.1007/s10533-006-9002-1>.

- Baigorría, T., Cazorla, C., Santos Sbuscio, D., Aimetta, B., Belluccini, P., 2012. Efecto de triticale ($\times$ Triticosecale Wittman) roloado como cultivo de cobertura en la supresión de malezas, rendimiento y margen bruto de soja. Informe de Actualización Técnica, EEA Marcos Juárez, p. 25.
- Baigorría, T., Alvarez, C., Cazorla, C., Belluccini, P., Aimetta, B., Pegoraro, V., Boccolini, M., Conde, B., Faggioli, V., Ortiz, J., Tuesca, D., 2019. Environmental impact and cover crop roller crimper in no-till soybean production | Impacto ambiental y roloado de cultivos de cobertura en producción de soja bajo siembra directa. *Ciencia Del Suelo* 37 (2).
- Barnes, A.P., Thomson, S.G., 2014. Measuring progress towards sustainable intensification: how far can secondary data go? *Ecol. Indic.* 36 <https://doi.org/10.1016/j.ecolind.2013.07.001>.
- Baulcombe, D., Crute, I., Davies, B., Dunwell, J., Gale, M., Jones, J., Prtty, J., Sutherland, W., Toulmin, C., 2009. Reaping the benefits: science and the sustainable intensification of global agriculture. Royal Society, London. ISBN: 978-0-85403-784-1. <http://royalsociety.org/uploadedFiles/Royal_Society_Content/policy/publications/2009/4294967719.pdf>. ISBN: 978-0-85403-784-1.
- Berhongaray, G., Alvarez, R., De Paepe, J., Caride, C., Cantet, R., 2013. Land use effects on soil carbon in the Argentinean pampas. *Geoderma* 192 (1). <https://doi.org/10.1016/j.geoderma.2012.07.016>.
- Bernardos, J., Zaccagnini, M.E., 2011. El uso de insecticidas en cultivos agrícolas y su riesgo potencial para las aves en la Región Pampeana. *El Hornero* 26 (1).
- Bojanich, M., Baigorría, T., Lardone, A., Cazorla, C., 2010. Análisis económico del cultivo de maíz sobre vicia utilizada como cultivo de cobertura. Informe técnico, EEA INTA Marcos Juárez.
- Bolinder, M.A., Janzen, H.H., Gregorich, E.G., Angers, D.A., VandenBygaart, A.J., 2007. An approach for estimating net primary productivity and annual carbon inputs to soil for common agricultural crops in Canada. *Agric. Ecosyst. Environ.* <https://doi.org/10.1016/j.agee.2006.05.013>.
- Brans, J.P., Vincke, P., 1985. Note—a preference ranking organisation method: (the PROMETHEE method for multiple criteria decision-making). *Manag. Sci.* 31 (6), 647–656. <https://doi.org/10.1287/mnsc.31.6.647>.
- Brans, J.-P., Vincke, P., Mareschal, B., 1986. How to select and how to rank projects: the PROMETHEE method. *Eur. J. Oper. Res.* 24 (2), 228–238. [https://doi.org/10.1016/0377-2217\(86\)90044-5](https://doi.org/10.1016/0377-2217(86)90044-5).
- Carciochi, W., Divito, G., Reussi, N., Echeverría, H., 2015. Las mejores prácticas de manejo de la fertilización azufrada en cultivos extensivos. *Informaciones Agronómicas de Hispanoamérica* 20, 2–6.
- Caride, C., Piñero, G., Paruelo, J.M., 2013. How does agricultural management modify ecosystem services in the Argentinean pampas? The effects on soil C dynamics. *Agric. Ecosyst. Environ.* 154, 23–33. <https://doi.org/10.1016/j.agee.2011.05.031>.
- Carnelos, D.A., Portela, S.I., Jobbágy, E.G., Jackson, R.B., Di Bella, C.M., Panario, D., Fagúndez, C., Piñero-Guerra, J.M., Grion, L., Piñero, G., 2019. A first record of bulk atmospheric deposition patterns of major ions in southern South America. *Biogeochemistry* 144, 261–271.
- Cassman, K.G., 1999. Ecological intensification of cereal production systems: yield potential, soil quality, and precision agriculture. *Proc. Natl. Acad. Sci. U. S. A.* 96, 5952–5959. <https://doi.org/10.1073/pnas.96.11.5952>.
- Caviglia, O.P., Andrade, F.H., 2010. Sustainable intensification of agriculture in the Argentinean pampas: capture and use efficiency of environmental resources. *Am. J. Plant Sci. Biotechnol.* 3, 1–8.
- Caviglia, O.P., Sadras, V.O., Andrade, F.H., 2004. Intensification of agriculture in the south-eastern pampas: I. capture and efficiency in the use of water and radiation in double-cropped wheat-soybean. *Field Crop Res.* 87 (2–3), 117–129. <https://doi.org/10.1016/j.fcr.2003.10.002>.
- Caviglia, O.P., Wingeyer, A.B., Novelli, L.E., 2016. El rol de los suelos agrícolas frente al cambio climático. *Serie de Extensión INTA Paraná* 78, 27–32.
- Collino, D.J., Salvaggiotti, F., Peticari, A., Piccinetti, C., Ovando, G., Urquiga, S., Racca, R.W., 2015. Biological nitrogen fixation in soybean in Argentina: relationships with crop, soil, and meteorological factors. *Plant Soil* 392, 239–252.
- Conway, G., Waage, J., 2010. Science and innovation for development science and innovation. *Society* 82 (3).
- Crespo, C., Wyngaard, N., Sainz Rozas, H., Studdert, G., Barraco, M., Gudelj, V., Barbagetala, P., Barbieri, P., 2021. Effect of the intensification of cropping sequences on soil organic carbon and its stratification ratio in contrasting environments. *Catena* 200. <https://doi.org/10.1016/j.catena.2021.105145>.
- Díaz-Balteiro, L., Iglesias-Merchan, C., Romero, C., de Jalón, S.G., 2020. The sustainable management of land and fisheries resources using multicriteria techniques: a meta-analysis. *Land* 9 (10). <https://doi.org/10.3390/land9100380>.
- Ding, G., Liu, X., Herbert, S., Novak, J., Amarasiwardena, D., Xing, B., 2006. Effect of cover crop management on soil organic matter. *Geoderma* 130 (3–4), 229–239. <https://doi.org/10.1016/j.geoderma.2005.01.019>.
- Doré, T., Makowski, D., Malézieux, E., Munier-Jolain, N., Tchamitchian, M., Titttonell, P., 2011. Facing up to the paradigm of ecological intensification in agronomy: revisiting methods, concepts and knowledge. *Eur. J. Agron.* <https://doi.org/10.1016/j.eja.2011.02.006>.
- Eggleston, H.S., Miwa, K., Srivastava, N., Iges, Tanabe K., 2006. IPCC 2008. 2006 IPCC Guidelines for National Greenhouse Inventories – A Primer, Prepared by the National Greenhouse gas Inventories Programme.
- Enrico, J.M., Piccinetti, C.F., Barraco, M.R., Agosti, M.B., Ecclesia, R.P., Salvaggiotti, F., 2020. Biological nitrogen fixation in field pea and vetch: response to inoculation and residual effect on maize in the Pampean region. *Eur. J. Agron.* 115, 126016.
- Evía, G., Sarandón, S.J., 2002. Aplicación del método multicriterio para valorar la sustentabilidad de diferentes alternativas productivas en los humedales de la Laguna Merín, Uruguay. *El Camino Hacia Una Agricultura Sustentable, Agroecología*.
- FAO, UNDP and UNEP, 2021. A multi-billion-dollar opportunity – Repurposing agricultural support to transform food systems. FAO, Rome. <https://doi.org/10.4060/cb6562en>.
- Ferraris, G.N., Elias, G., Galetto, M.L., 2014. Pérdidas de nitrógeno por volatilización en maíces tardíos en Pergamino (BS AS). Efectos de fuente y dosis. *Actas X Congreso Nacional de maíz*. Rosario, Santa Fe, pp. 3–5.
- Ferraro, D.O., Ghersa, C.M., Sznajder, G.A., 2003. Evaluation of environmental impact indicators using fuzzy logic to assess the mixed cropping systems of the inland Pampa, Argentina. *Agric. Ecosyst. Environ.* 96 (1–3) [https://doi.org/10.1016/S0167-8809\(03\)00017-3](https://doi.org/10.1016/S0167-8809(03)00017-3).
- Firbank, L.G., Elliott, J., Drake, B., Cao, Y., Gooday, R., 2013. Evidence of sustainable intensification among British farms. *Agric. Ecosyst. Environ.* 173 <https://doi.org/10.1016/j.agee.2013.04.010>.
- Foley, J.A., Ramankutty, N., Brauman, K.A., Cassidy, E.S., Gerber, J.S., Johnston, M., et al., 2011. Solutions for a cultivated planet. *Nature* 478 (7369). <https://doi.org/10.1038/nature10452>.
- Gadanakis, Y., Bennett, R., Park, J., Areal, F.J., 2015. Evaluating the sustainable intensification of arable farms. *J. Environ. Manag.* 150 <https://doi.org/10.1016/j.jenvman.2014.10.005>.
- García, F., González Sanjuan, M.F., 2010. Balances de nutrientes en Argentina ¿Cómo estamos? ¿Cómo mejoramos? *Informaciones Agronómicas del Cono Sur* 48 (6).
- Garnett, T., Godfray, C., 2012. Sustainable intensification in agriculture. Navigating a course through competing food system priorities. In: *Food Climate Research Network and the Oxford Martin Programme on the Future of Food* (July).
- Ghida Daza, C., 2009. Indicadores económicos para la gestión de empresas agropecuarias. Ediciones INTA ISSN 1851-6955 N° 11.
- Godfray, H.C.J., Beddington, J.R., Crute, I.R., Haddad, L., Lawrence, D., Muir, J.F., Toulmin, C., 2010. Food security: the challenge of feeding 9 billion people. *Science*. <https://doi.org/10.1126/science.1185383>.
- Gregorich, E.G., Rochette, P., St-Georges, P., McKim, U.F., Chan, C., 2008. Tillage effects on N2O emission from soils under corn and soybeans in eastern Canada. *Can. J. Soil Sci.* 88 (2) <https://doi.org/10.4141/CJSS06041>.
- Hatch, D.K., Goulding, W.T., Murphy, D.V., 2003. Nitrogen. In: Haygarth, P.M., Jarvis, S. C. (Eds.), *Agriculture, Hydrology and Water Quality*. CABI Publishing, Wallingford, pp. 7–27.
- Hisse, I.R., Biganzoli, F., Peper, A.M., Poggio, S.L., 2022. Annual productivity of cropping sequences: responses to increased intensification levels. *Eur. J. Agron.* 137, 126506. <https://doi.org/10.1016/j.eja.2022.126506>.
- Kovach, J., Peltzold, C., Degni, J., Tette, J., 1992. A method to measure the environmental impact of pesticides. *New York's Food Life Sci. Bull.* 139 (139).
- Król, A., Książek, J., Kubińska, E., Rozakis, S., 2018. Evaluation of sustainability of maize cultivation in Poland. A prospect theory-PROMETHEE approach. *Sustainability* (Switzerland) 10 (11). <https://doi.org/10.3390/su10114263>.
- Kumar, P., Kumar, S., 2018. Agricultural diversification – an opportunity for smallholders (A case study of Sonapat District of Haryana). *IOSR J. Hum. Soc. Sci.* 23 (1), India.
- Kyili, A., Christoforou, E., Fokaidis, P.A., y Polycarpou, P., 2016. Multicriteria analysis for the selection of the most appropriate energy crops: the case of Cyprus. *Int. J. Sustain. Energy* 35 (1).
- Lewczuk, N.A., 2017. Estudio de las emisiones de óxido nítrico en sistemas agrícolas de la región pampeana. Facultad de Ciencias Exactas y Naturales. Universidad de Buenos Aires.
- Macharis, C., Springael, J., De Brucker, K., Verbeke, A., 2004. PROMETHEE and AHP: the design of operational synergies in multicriteria analysis - strengthening PROMETHEE with ideas of AHP. In: *Eur. J. Oper. Res.* 153 [https://doi.org/10.1016/S0377-2217\(03\)00153-X](https://doi.org/10.1016/S0377-2217(03)00153-X).
- Mahon, N., Crute, I., Di Bonito, M., Simmons, E.A., Islam, M.M., 2018. Towards a broad-based and holistic framework of sustainable intensification indicators. *Land Use Policy* 77. <https://doi.org/10.1016/j.landusepol.2018.06.009>.
- Mareschal, B., De Smet, Y., 2009. Visual PROMETHEE: Developments of the PROMETHEE & GAIA multicriteria decision aid methods. In: In 2009 IEEE International Conference on Industrial Engineering and Engineering Management. IEEE, pp. 1646–1649.
- Martínez, J.P., Crespo, C., Sainz Rozas, H., Echeverría, H., Studdert, G., Martínez, F., Barbieri, P., 2020. Carbono orgánico del suelo en secuencias de cultivo con predominio de soja en la Pampa húmeda argentina. *Manejo y uso del suelo* 36 (1), 173–183.
- Mary, B., Wylleman, R., 2001. Characterization and modelling of organic C and N in soil in different cropping systems. In: *Proceedings of the 11th Nitrogen Workshop*. Reims, France, pp. 251–252.
- Milesi Delaye, L.A., Irizar, A.B., Andrioli, A.E., Mary, B., 2013. Effect of continuous agriculture of grassland soils of the Argentine Rolling Pampa on soil organic carbon and nitrogen. *Appl. Environ. Soil Sci.* 2013 <https://doi.org/10.1155/2013/487865>.
- Mouratiadou, I., Latka, C., van der Hilst, F., Müller, C., Berges, R., Bodirsky, B.L., Ewert, F., Faye, B., Heckelet, T., Hoffman, M., Lehtonen, H., Lortie, I.J., Nende, C., Palosuo, T., Rodríguez, A., Rotter, R.P., Ruiz-Ramos, M., Stella, T., Webber, H., Wicke, B., 2021. Quantifying sustainable intensification of agriculture: the contribution of metrics and modelling. *Ecol. Indic.* 129 <https://doi.org/10.1016/j.ecolind.2021.107870>.
- Oenema, O., Kros, H., De Vries, W., 2003. Approaches and uncertainties in nutrient budgets: implications for nutrient management and environmental policies. In: *Eur. J. Agron.* 20 [https://doi.org/10.1016/S1161-0301\(03\)00067-4](https://doi.org/10.1016/S1161-0301(03)00067-4).
- Parkin, T.B., Kaspar, T.C., 2006. Nitrous oxide emissions from corn-soybean systems in the Midwest. *J. Environ. Qual.* 35 (4) <https://doi.org/10.2134/jeq2005.0183>.
- Parkin, M., Loria, E., 2010. *Microeconomía de Parkin*. Pearson.

- Pecar, M., 2008. Teoría de Portafolio: Utilización para Evaluar los Riesgos Agropecuarios. Oficina de Riesgo Agropecuario de la Secretaría de Agricultura, Ganadería, Pesca y Alimentos.
- Piccinetti, C.F., Bacigaluppo, S., Di Ciocco, C.A., De Tellería, J.M., Salvagioti, F., 2021. Soybean in rotation with cereals attenuates nitrous oxide emissions as compared with soybean monoculture in the pampas region. *Geoderma* 402, 115192.
- Poeplau, C., Don, A., 2015. Carbon sequestration in agricultural soils via cultivation of cover crops - a meta-analysis. *Agric. Ecosyst. Environ.* <https://doi.org/10.1016/j.agee.2014.10.024>.
- Ponce, C., Arnillas, C., y Escobal, J., 2015. Cambio climático, uso de riego y estrategias de diversificación de cultivos en la sierra peruana. Nuevas miradas desde el Censo Agropecuario. Grade, Perú, En Agricultura Peruana.
- Portela, S.I., Restovich, S.B., González, H.M., Torti, M.J., 2016. Deep drainage and nitrogen leaching reduction in crop rotations with cover crops. *Ecol. Aust.* 26 (3). http://hdl.handle.net/20.500.12110/ecologiaaustral_v026_n03_p212.
- Pretty, J., 2008. Agricultural sustainability: concepts, principles and evidence. *Philos. Trans. Royal Soc. B Biol. Sci.* <https://doi.org/10.1098/rstb.2007.2163>.
- Pretty, J., Bharucha, Z.P., 2014. Sustainable intensification in cropping systems. *Ann. Bot.* <https://doi.org/10.1093/aob/mcu205>.
- Pretty, J., Toulmin, C., Williams, S., 2011. Sustainable intensification in African agriculture. *Int. J. Agric. Sustain.* 9 (1) <https://doi.org/10.3763/ijas.2010.0583>.
- Pretty, J., Benton, T.G., Bharucha, Z.P., Dicks, L.V., Flora, C.B., Godfray, H.C.J., Goulson, D., Hartley, S., Lampkin, N., Morris, C., Pierzynski, G.P., Prasad, V.V., Reganold, J., Rockström, J., Smith, P., Thorne, P., Wratten, S., 2018. Global assessment of cropping system redesign for sustainable intensification. *Nat. Sustain.* 1 (8) <https://doi.org/10.1038/s41893-018-0114-0>.
- Principiano, M., Acciaresi, H.A., 2017. Costo financiero e impacto ambiental del control de malezas en diferentes secuencias de cultivos en el NO de la Provincia de Buenos Aires. *RTA / Vol 10 / N°33: 37–4*.
- Principiano, M.A., Acciaresi, H.A., 2018. Estado de situación del uso de herbicidas en cultivos extensivos agrícolas de la región noroeste de la provincia de Buenos Aires y el impacto ambiental asociado. Ediciones INTA.
- PROMETHEE, Visual, 2013. Visual PROMETHEE 1.4 manual.
- Restovich, S.B., Andriulo, A.E., Armas-Herrera, C.M., Beribe, M.J., Portela, S.I., 2019. Combining cover crops and low nitrogen fertilization improves soil supporting functions. *Plant Soil* 442 (1–2). <https://doi.org/10.1007/s11104-019-04205-8>.
- Ritchie, S.W., Hanway, J.J., Benson, G.O., 1992. How a plant crop develops. Iowa State University of Science and Technology, Coop. Ext. Serv, Ames, Iowa, USA.
- Rockström, J., Williams, J., Daily, G., Noble, A., Matthews, N., Gordon, L., 2017. Sustainable intensification of agriculture for human prosperity and global sustainability. *Ambio* 46 (1). <https://doi.org/10.1007/s13280-016-0793-6>.
- Rodríguez, M.P., Domínguez, A., Moreira Ferroni, M., Wall, L.G., Bedano, J.C., 2020. The diversification and intensification of crop rotations under no-till promote earthworm abundance and biomass. *Agronomy* 10 (7), 919. <https://doi.org/10.3390/agronomy10070919>.
- Rzezak, H.F., 2008. El conflicto entre el Gobierno y el campo en Argentina. Lineamientos políticos, estrategias discursivas y discusiones teóricas a partir de un abordaje multidisciplinar. *Iberoforum. Revista de Ciencias Sociales de la Universidad Iberoamericana* 3 (6), 82–106.
- Sadok, W., Angevin, F., Bergez, J.E., Bockstaller, C., Colomb, B., Guichard, L., Reau, R., Doré, T., 2008. Ex ante assessment of the sustainability of alternative cropping systems: implications for using multi-criteria decision-aid methods. A review. *Agron. Sustain. Dev.* <https://doi.org/10.1051/agro:2007043>.
- Shukla, P., Skea, J., Buendia, E.C., 2019. IPCC, 2019: Climate Change and Land: An IPCC Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food. IPCC.
- Smith, A., Snapp, S., Chikowo, R., Thorne, P., Bekunda, M., Glover, J., 2017. Measuring sustainable intensification in smallholder agroecosystems: a review. *Glob. Food Secur.* <https://doi.org/10.1016/j.gfs.2016.11.002>.
- Ministry of Agriculture, Argentina, 2022. <https://public.tableau.com/app/profile/fhorm/viz/Cultivosdecoberturav2/Dashboard>. Accessed in 12 February 2022.
- Talukder, B., 2016. Multi-Criteria Decision Analysis (MCDA) for Agricultural Sustainability Assessment.
- Talukder, B., Hipel, K.W., 2018. The PROMETHEE framework for comparing the sustainability of cropping systems. *Resources* 7 (4). <https://doi.org/10.3390/resources7040074>.
- Tittonell, P., 2014. Ecological intensification of agriculture-sustainable by nature. *Curr. Opin. Environ. Sustain.* <https://doi.org/10.1016/j.cosust.2014.08.006>.
- Vancley, F., 2009. Multifunctional agriculture: a transition theory perspective - by Geoff A. Wilson. *Geogr. Res.* 47 (2) <https://doi.org/10.1111/j.1745-5871.2009.00580.x>.
- Vicente, G., Engler, P., 2008. Valoración económica del balance de nitrógeno y fósforo de los principales rubros agrícolas y pecuarios en la provincia de Entre Ríos. *Agricultura Sustentabl Serie Extensión* 51.
- Wander, M.M., Traina, S.J., 1996. Organic matter fractions from organically and conventionally managed soils: I. Carbon and nitrogen distribution. *Soil Sci. Soc. Am. J.* 60 (4), 1081–1087. <https://doi.org/10.2136/sssaj1996.03615995006000040017x>.
- Weltin, M., Zasada, I., Piore, A., Debolini, M., Geniaux, G., Moreno, O., Scherer, L., Lorena, Tudela M., Schulp, C.J.E., 2018. Conceptualising fields of action for sustainable intensification – A systematic literature review and application to regional case studies. *Agric. Ecosyst. Environ.* 257, 68–80.
- Wingeyer, A.B., Amado, T.J.C., Pérez-Bidegain, M., Studdert, G.A., Perdomo Varela, C. H., García, F.O., Karlen, D.L., 2015. Soil quality impacts of current South American agricultural practices. *Sustainability* (Switzerland). <https://doi.org/10.3390/su7022213>.